`bispectrum`: Selective G-Bispectra Made Practical

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Abstract

Many machine-learning tasks are invariant under group actions: image classification under planar rotation $\mathrm{SO}(2)$, signal classification under translation, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. The G-bispectrum is a principled *complete* invariant (it preserves all signal information up to the group action) but its deployment has been hampered by quadratic cost and a patchwork of group-specific implementations. We present `bispectrum`, an open-source, fully unit-tested PyTorch library that computes *selective* G-bispectral invariants for seven different group/domain pairs as drop-in differentiable `nn.Modules`. Selectivity reduces the invariant size from $O(|G|^2)$ for the full bispectrum to $O(|G|)$ while remaining complete on generic signals. On the theoretical side, we contribute an augmented selective $\mathrm{SO}(3)$ invariant on the sphere S^2 that achieves $\Theta(L^2)$ coefficients at band-limit L (matching the information-theoretic lower bound) instead of the $O(L^3)$ of the full bispectrum, with a computer-assisted completeness proof for $4 \leq L \leq 100$. We benchmark the library on three datasets across three group/domain pairs — PCam under C_8, OrganMNIST3D under the octahedral group, and Spherical MNIST under $\mathrm{SO}(3)$ — comparing against norm pooling, gated pooling, Fourier-ELU pooling, max pooling, and (non-equivariant) augmented CNN baselines. The library delivers near-exact rotation invariance (test-set rotation-accuracy std $\sigma_{\mathrm{rot}} < 10^{-3}$) and a data-efficiency advantage over incomplete alternatives in the low-data, moderate-capacity regime; the selective forward pass runs in sub-millisecond time on GPU for finite groups. Code is available at `https://github.com/geometric-intelligence/bispectrum`.

1 Introduction

Many machine-learning tasks are invariant under the action of a symmetry group G: histopathology classification under planar rotations C_n, signal classification under translations, volumetric organ classification under cube rotations O, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. Equivariant neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] enforce these symmetries by construction in their feature maps, but at the point where a prediction must be made any task that does not depend on the group action — classification, regression, detection — still has to map the equivariant features to an *invariant* representation.

The *G-bispectrum* is a complex-valued vector of triple-correlation coefficients of the group Fourier transform. It has three properties that matter for ML: (i) **completeness** — it preserves all signal information up to the group action (the map is one-to-one on group orbits); (ii) **inversion** — the signal can in principle be recovered from the invariant up to its orbit; and (iii) **selectivity** — a small $O(|G|)$ index subset of the $O(|G|^2)$ full bispectrum still suffices for completeness on generic signals [Mataigne et al., 2024].

Existing invariant pooling layers — norm pooling, gate-based pooling, max/sum aggregation [Weiler and Cesa, 2019] — are simple but *incomplete*: they collapse equivariant features to scalars and dis-

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card information that distinguishes signals in different orbits. The bispectrum offers a mathematically principled alternative [Kakarala, 2012] and has been used as a competitive pooling layer in equivariant networks [Sanborn et al., 2023, Mataigne et al., 2024].

Despite this, bispectral pooling has not been widely adopted, for both theoretical and computational reasons. Existing bispectral constructions are scattered across theory papers, domain-specific derivations, and small implementations, with no general-purpose PyTorch library covering the discrete and continuous symmetry groups relevant to modern geometric deep learning. Even for groups where selectivity [Mataigne et al., 2024] reduces the bispectrum from $O(|G|^2)$ to $O(|G|)$ scalars, a user still needs a numerically stable implementation, a consistent API, an inversion routine, tests, and benchmarked runtime. On the theoretical side, the selective construction of Mataigne et al. [2024] applies to finite groups; the continuous case of $\mathrm{SO}(3)$ acting on the sphere S^2 has remained open.

This paper presents `bispectrum`, a PyTorch [Paszke et al., 2019] library for computing bispectral invariants and, where available, their inverses. The library supports cyclic, dihedral, torus, disk, sphere, and octahedral settings as differentiable `nn.Modules` intended to plug into standard ML workflows. Beyond the software contribution, the paper makes one technical contribution that arose during implementation: an augmented selective $\mathrm{SO}(3)$ bispectrum on S^2 with $\Theta(L^2)$ coefficients (Section 3.3).

Contributions.

1. **Software contribution.** A PyTorch library implementing selective G-bispectra for *seven different group/domain pairs* (C_n on C_n, $\mathrm{SO}(2)$ on S^1, $\mathbb{T}^d$, D_n, $\mathrm{SO}(2)$ on the disk, $\mathrm{SO}(3)$ on S^2, and the octahedral group O) as drop-in differentiable `nn.Modules` with a consistent API, GPU-optimised forward passes ($<$1 ms for finite groups), precomputed CG buffers, and a fully unit-tested codebase (96.8% line coverage).
2. **Theoretical contribution: augmented selective $\mathrm{SO}(3)$ bispectrum on S^2.** We construct an invariant on S^2 that uses $\Theta(L^2)$ scalar coefficients (matching the dimension of the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ up to a constant factor) instead of the $O(L^3)$ of the full bispectrum. The invariant augments the scalar bispectrum with degree-4 *CG power spectrum* entries $P_{\ell_1,\ell_2,\ell} = \|(\hat{f}_{\ell_1} \otimes \hat{f}_{\ell_2})|_\ell\|^2$ that are needed to restore rank for real-valued signals. We give a computer-assisted proof of completeness for every band-limit $4 \leq L \leq 100$, combining a structural seed step [Kakarala, 1992, 2012] with per-degree linear-system rank certificates verified at deterministic witnesses; the verification script is uniform in L and extendable to larger band-limits without changes to the structural argument (Section 3.3, Appendix A).
3. **Empirical validation.** A benchmark of G-bispectral pooling across three ML tasks — PCam under C_8, OrganMNIST3D under the octahedral group O, and Spherical MNIST under $\mathrm{SO}(3)$ — against norm pooling, gated pooling, Fourier-ELU pooling, max pooling, and (non-equivariant) augmented-CNN baselines. Complete bispectral invariants provide a data-efficiency advantage in the low-data, moderate-capacity regime while converging to incomplete alternatives at high capacity.

Together, these contributions provide the first general-purpose, drop-in differentiable library for G-bispectral invariants and the first practical extension of the selective construction from finite groups to $\mathrm{SO}(3)$ on S^2.

2 Mathematical Background

Notation. A *group* G is a set with an associative product, identity, and inverse (e.g. rotations of the plane, translations of an image grid). G is *abelian* if elements commute (e.g. cyclic groups C_n, the torus $\mathbb{T}^d$) and *non-abelian* otherwise (e.g. $\mathrm{SO}(3)$, the dihedral groups D_n, the octahedral group O). A *representation* ρ assigns to each $g \in G$ a unitary matrix $\rho(g)$ such that $\rho(gh) = \rho(g)\rho(h)$; an *irreducible representation* (*irrep*) admits no nontrivial invariant subspace. The *dual* $\hat{G}$ denotes the set of equivalence classes of irreps; for finite G, $|\hat{G}| = O(|G|)$. A *group action* of G on a domain X is a map $G \times X \to X$, $(g, x) \mapsto g \cdot x$, compatible with the group product; it induces an action on functions $f : X \to \mathbb{R}$ by $(f \circ g)(x) := f(g \cdot x)$. Two signals are in the same *orbit* if one is the image

Table 1: Invariant nonlinearities for equivariant CNNs. "Exact"* means the invariant has no discretisation/aliasing error; "Complete" means it separates orbits; the coefficient column counts the size of the output for a signal with $|G|$ Fourier samples. "Genericity assumption"† means completeness holds outside a Lebesgue-measure-zero algebraic subvariety of the signal space. The selective bispectrum is the only method that is simultaneously exact, complete, and linear-cost for finite groups.

Nonlinearity	Exact?*	Complete?	Coefficients	Key limitation
Norm	Yes	No	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(2\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes‡	$O(\|G\| \log \|G\|)$	Aliasing
Full bispectrum	Yes	Yes	$O(\|G\|^2)$	Quadratic cost
Selective bispectrum	**Yes**	**Yes**	$\mathbf{O(\|G\|)}$	Genericity assumption†

‡Complete pre-aliasing; the pointwise nonlinearity in the forward pass introduces aliasing that may break completeness in practice.

of the other under some $g \in G$; an *invariant* of f is a function constant on each orbit. For compact G there is a unique normalised translation-invariant measure $d\mu$ called the *Haar measure*; finite G is the special case where $d\mu$ is the counting measure. "*Lebesgue-almost-every* f" means "every f outside a measure-zero algebraic subvariety of the signal space." The groups and domains used in this paper are: the cyclic group C_n (discrete planar rotations), the dihedral group D_n (rotations and reflections of a regular n-gon), the torus $\mathbb{T}^d = (S^1)^d$ (multi-channel periodic signals), $\mathrm{SO}(2)$ acting on the disk (continuous planar rotations of a 2D image), $\mathrm{SO}(3)$ acting on the sphere S^2 (continuous 3D rotations of a spherical signal), and the chiral octahedral group O (24 proper rotations of the cube, acting on 3D voxel grids).

2.1 Equivariant features and invariant maps

A neural network layer Φ is *G-equivariant* if its output transforms predictably under the group action: $[\pi_{\mathrm{out}}(g) \circ \Phi](f) = \Phi(\pi_{\mathrm{in}}(g)f)$ for all $g \in G$ [Cohen and Welling, 2016, Weiler and Cesa, 2019]. Here f is an input signal (a function on the domain X), $\pi_{\mathrm{in}}, \pi_{\mathrm{out}}$ are the (representations of the) G-actions on the input and output feature spaces of Φ, and the composition $\pi_{\mathrm{out}}(g) \circ \Phi$ is the action of g on the output side. Stacking equivariant layers preserves equivariance, but any task that requires a prediction independent of the group action—classification, regression, detection—must eventually map equivariant features to *invariant* ones (constant on each orbit).

Table 1 summarises the trade-offs between exactness, completeness, and computational cost across existing invariant pooling layers. Norm pooling and gated nonlinearities are exact invariants but *incomplete*: they discard the phase of Fourier blocks, collapsing distinct orbits. Fourier-space pointwise nonlinearities preserve completeness but sacrifice exactness through aliasing [Weiler and Cesa, 2019]. The full bispectrum is exact and complete but costs $O(|G|^2)$ coefficients. The *selective* bispectrum (introduced below) preserves both exactness and completeness while using only $O(|G|)$ coefficients on generic signals.

2.2 The G-bispectrum and selective bispectra

Let G be a compact group with dual $\hat{G}$ and irreps $\{\rho_i\}_{i \in \hat{G}}$ of dimensions d_i. The *group Fourier transform* of a signal $f : G \to \mathbb{C}$ is

$$\hat{f}(\rho_i) = \int_G f(g)\, \rho_i(g)\, d\mu(g) \quad \in \mathbb{C}^{d_i \times d_i}, \tag{1}$$

which specialises to $\sum_{g \in G} f(g)\, \rho_i(g)$ when G is finite (counting measure). The Clebsch–Gordan (CG) decomposition $\rho_i \otimes \rho_j \cong \bigoplus_{k \in \hat{G}} m_{ij}^k \rho_k$ expresses the tensor product of two irreps as a direct sum of irreps (with integer multiplicities $m_{ij}^k \geq 0$); the change-of-basis is realised by the *CG isometry* $C_{ij}^k : V_{\rho_k} \to V_{\rho_i} \otimes V_{\rho_j}$, projecting the tensor product onto its ρ_k-isotypic component. For each admissible triple (i, j, k) with $m_{ij}^k > 0$, the *bispectrum* is the scalar

$$\beta_{i,j}^k(f) := \big\langle\, (C_{ij}^k)^\dagger \big(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j)\big),\, \hat{f}(\rho_k) \big\rangle. \tag{2}$$

Table 2: Current scope of the `bispectrum` library. `SO2onS1` is the continuous-n limit of `CnonCn` and shares its implementation.

Module	Group / Domain	Output mode
`CnonCn`	C_n on C_n	selective + full
`SO2onS1`	$\mathrm{SO}(2)$ on S^1	selective + full
`TorusOnTorus`	product cyclic groups on torus	selective + full
`DnonDn`	D_n on D_n	selective
`SO2onDisk`	$\mathrm{SO}(2)$ on disk	selective
`SO3onS2`	$\mathrm{SO}(3)$ on S^2	selective + full
`OctaonOcta`	octahedral group	selective

Each $\beta^k_{i,j}$ is invariant under the action $f \mapsto f \circ g$ and, taken over all admissible triples, the bispectrum $\{\beta^k_{i,j}\}_{(i,j,k)}$ is a *complete* invariant: it separates orbits for generic signals [Kakarala, 2012]. For finite G, $|\hat{G}| = O(|G|)$, so the full bispectrum has $O(|G|^2)$ scalar coefficients (and is naturally indexed by $(i,j) \in \hat{G}^2$ with k ranging over the CG-allowed irreps). Equation (2) is stated for G acting on itself; for homogeneous spaces of the form G/H (e.g. $\mathrm{SO}(3)$ on S^2, or $\mathrm{SO}(2)$ on the disk), the same construction applies after expanding f in a basis adapted to G/H (spherical harmonics for S^2, Bessel modes on the disk). The explicit specialisation for S^2 is in Definition 5.

Selective bispectra. Mataigne et al. [2024] show that for finite groups, a breadth-first search (BFS) over the Kronecker product table of $\hat{G}$ selects $O(|G|)$ bispectral coefficients that remain complete; Myers and Miolane [2025] extend this to the disk. This reduction from quadratic to linear is the key to practical deployment: for example, the bispectrum of the octahedral group drops from 576 to 172 coefficients.

3 The `bispectrum` Library

3.1 Design and implementation

Every group/domain pair is exposed as a `torch.nn.Module` named `{Group}on{Domain}` (e.g. `SO3onS2`). Each module exposes a uniform API: `forward(f)` returns the selective bispectral invariants of f; `fourier(f)` returns the underlying group Fourier coefficients; and, where available, `invert(beta)` returns the reconstructed signal up to group-action indeterminacy. Modules accept raw real-valued inputs, handle the Fourier transform internally, and default to $O(|G|)$ selective coefficients (`selective=False` gives the full $O(|G|^2)$ set). Clebsch–Gordan (CG) matrices, discrete Fourier transform (DFT) kernels, and Bessel roots (used for the radial discretisation of the disk in `SO2onDisk`) are precomputed at construction and stored as non-learnable buffers; this lets `.to(device)` migrate the module transparently and ensures that forward passes incur zero recomputation. The only dependencies are PyTorch [Paszke et al., 2019], NumPy [Harris et al., 2020], and `torch_harmonics` [Bonev et al., 2023] (for `SO3onS2`). For $\mathrm{SO}(3)$ on S^2, reduced CG matrices are built in parallel and cached to disk; all forward methods are batch-native, using `torch.einsum`/`torch.bmm` for non-abelian modules and `torch.fft` for abelian ones. The test suite covers 96.8% of lines across all modules.

3.2 Supported modules

Example API. A minimal usage example is in Appendix G.

3.3 Augmented selective $\mathrm{SO}(3)$ bispectrum on S^2

A real-valued signal on the sphere $f : S^2 \to \mathbb{R}$ that is *band-limited at degree* L expands in spherical harmonics as $f(\theta,\varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m Y_\ell^m(\theta,\varphi)$ (Eq. 3), so it is parameterised by $(L+1)^2$ real scalars (one per (ℓ,m) pair, after the reality constraint $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$). The full bispectrum on S^2 at band-limit L has $\Theta(L^3)$ scalar entries $\beta_{\ell_1,\ell_2,\ell}$ ($0 \le \ell_1 \le \ell_2 \le L$, $|\ell_1 - \ell_2| \le \ell \le \min(\ell_1+\ell_2, L)$), the specialisation of Eq. 2 to the $\mathrm{SO}(3)$-on-S^2 case (Definition 5).

For finite groups, the selective construction of Mataigne et al. [2024] reduces $O(|G|^2)$ to $O(|G|)$ by a breadth-first traversal of the irrep graph. On S^2 the analogous chain $\beta_{1,0,1}, \beta_{1,1,2}, \beta_{1,2,3}, \ldots, \beta_{1,L-1,L}$ provides only L complex coefficients ($\sim 3L$ real scalars after accounting for parity), e.g. ~ 15 for $L = 5$ vs. $(L{+}1)^2 = 36$ unknowns. Indeed, the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L{+}1)^2 - \dim \mathrm{SO}(3) = (L{+}1)^2 - 3$, so any complete invariant requires $\Omega(L^2)$ independent components. We achieve this lower bound.

Theorem 1 (Augmented selective $\mathrm{SO}(3)$ invariant on S^2)**.** *For $f : S^2 \to \mathbb{R}$ band-limited at degree $4 \le L \le 100$, there exists an augmented invariant $\widetilde{\beta}_{\mathrm{aug}}(f)$ consisting of $\Theta(L^2)$ scalars—a subset $\mathcal{S}_\beta$ of scalar bispectral entries $\beta_{\ell_1,\ell_2,\ell}$ together with a subset $\mathcal{S}_P$ of CG power spectrum entries $P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2$—that is a complete $\mathrm{SO}(3)$-invariant for* Lebesgue-almost-every *signal (i.e. for every signal outside a measure-zero algebraic subvariety of the $(L{+}1)^2$-dimensional signal space). The output size $\Theta(L^2)$ matches the dimension $(L{+}1)^2 - 3$ of the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ up to a constant factor.*

Remark 2. The proof (Appendix A) is *algorithmic*: it shows completeness of $\widetilde{\beta}_{\mathrm{aug}}$ by reconstructing the spherical-harmonic coefficients $(\mathbf{F}_0, \ldots, \mathbf{F}_L)$ of f from the invariant, in four steps: (i) *gauge-fixing*, choosing a unique representative in the $\mathrm{SO}(3)$-orbit of f by aligning $\mathbf{F}_1$ with the z-axis and fixing the residual azimuthal phase (Lemma 18); (ii) *seed recovery* at low degrees $\ell \le L_0 = 4$, where adjoining a constant-size block of full bispectral entries lets us invoke Kakarala's [Kakarala, 1992, 2012] completeness theorem to determine $\mathbf{F}_0, \ldots, \mathbf{F}_4$ (Lemma 24); (iii) *per-degree linear-system inversion* for $\ell \ge 5$, where each $\mathbf{F}_\ell$ is recovered by solving an $O(\ell)$-dimensional linear system $A_\ell \mathbf{F}_\ell = b_\ell$ whose coefficients depend on already-recovered lower degrees; and (iv) *computer-algebra rank certificates* that verify $\det(A_\ell) \ne 0$ at deterministic witness points (seed 42) for each $\ell \le 100$, which propagates to a generic non-vanishing statement (the complement of a Lebesgue-null algebraic subvariety) by standard algebraic-variety arguments. The structural form of the linear system is uniform for $\ell \ge 8$; extending the verification beyond $L = 100$ requires only re-running a single script.

Why augmentation is necessary. The reality constraint $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ on the spherical-harmonic coefficients of a real signal is *not* a generic constraint — it cuts the complex parameter space along a measure-zero linear subvariety. Generic statements proved on the complex parameter space therefore need not transfer to the real subspace, and indeed they do not: at every degree $\ell \ge 4$, the per-degree linear system of *scalar* bispectral entries has full *complex* rank $2\ell{+}1$ on the complex parameter space (Lemma 32, verified at explicit witnesses up to $\ell = 100$), but its restriction to the real-signal slice has only partial *real Jacobian* rank $\sim \ell$. At $\ell = 4$, for example, the linear bispectral real Jacobian has rank 4 instead of $9 = 2\ell{+}1$, and this rank deficiency is intrinsic — it persists even when all triples of the *full* bispectrum are used and when the self-coupling entries $\beta_{\ell,\ell,\ell'}$ are added, so it cannot be resolved by choosing a different scalar subset. The CG power spectrum entries $P_{\ell_1,\ell_2,\ell}$ capture the complementary *magnitude* information that the phase-sensitive scalar bispectral entries miss on the real subspace, and together they achieve full real Jacobian rank $2\ell{+}1$ at every degree (Lemma 35).

The selective index set. The bispectral component $\mathcal{S}_\beta$ selects up to $2\ell{+}1$ triples per degree from four categories: **(1) Chain** (ℓ_1, ℓ_2, ℓ) with $\ell_1 \le \ell_2 < \ell$; **(2) Cross** (ℓ_1, ℓ, ℓ') with $\ell_1, \ell' < \ell$; **(3) Power** $(0, \ell, \ell)$; **(4) Self-coupling** (ℓ, ℓ, ℓ') with ℓ' even. Entries with $\ell_1 = \ell_2$ and odd ℓ vanish identically for real signals: the CG coefficients $\langle \ell, m_1; \ell, m_2 \mid \ell, m \rangle$ are antisymmetric under $m_1 \leftrightarrow m_2$ when the total parity $2\ell{+}\ell$ is odd, while the polynomial $a_\ell^{m_1} a_\ell^{m_2}$ is symmetric, so the contraction sums to zero (Proposition 29). Such entries are excluded. All *even* self-coupling entries $\beta_{\ell,\ell,\ell'}$ with $2 \le \ell' \le \ell$ are included unconditionally; they are needed for full Jacobian rank in the seed system (Appendix B). The invariant is further augmented with CG power spectrum entries $P_{\ell_1,\ell_2,\ell}$, chosen to maximise the Jacobian rank of the combined invariant (Algorithm 1).

Table 7 (Appendix A) gives the per-degree entry counts of $\widetilde{\beta}_{\mathrm{aug}}$ (scalar bispectral entries plus CG power scalars); at degree $\ell \ge 5$ the total is $\sim 3\ell$, confirming the $\Theta(L^2)$ output size.

Proof sketch. At a high level, we prove that $\widetilde{\beta}_{\mathrm{aug}}$ is complete by reconstructing the spherical-harmonic expansion of f from the invariant; once the coefficient vectors $(\mathbf{F}_0, \ldots, \mathbf{F}_L)$ are recovered, f itself follows by inverse spherical-harmonic transform. The reconstruction proceeds in two stages:

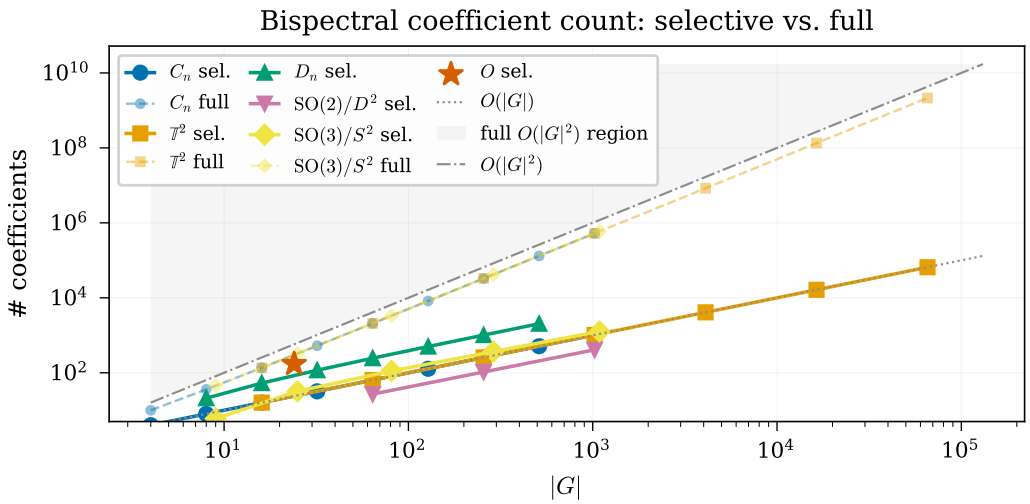

Figure 1: Bispectral coefficient count vs. group order $|G|$. Solid lines: selective ($O(|G|)$); dashed: full ($O(|G|^2)$). At $|G|$=1,024, selectivity yields a 512× reduction. Continuous groups (SO(2), SO(3)) are plotted at their band-limited signal dimension. The full-bispectrum curves for D_n and the disk are omitted because the corresponding backends are not yet implemented in the library.

a constant-size *seed* that recovers $\mathbf{F}_0, \ldots, \mathbf{F}_4$ from a $\Theta(1)$-block of adjoined full bispectral entries, then a degree-by-degree *linear-system inversion* that recovers each $\mathbf{F}_\ell$ ($\ell \geq 5$) from a per-degree linear system whose coefficients depend on the already-known lower degrees.

In more detail: after gauge-fixing ($\mathbf{F}_0$ from $\beta_{0,0,0} = (a_0^0)^3$, $\mathbf{F}_1$ from $\beta_{0,1,1}/a_0^0 = \|\mathbf{F}_1\|^2$ together with the z-axis alignment of Lemma 18), we adjoin the full bispectrum at $L_0 = 4$ ($\leq$30 extra scalars, constant in L and asymptotically free) and apply Kakarala's theorem [Kakarala, 2012, 1992] to recover $\mathbf{F}_0, \ldots, \mathbf{F}_4$ on the gauge slice; the orientation-reversing reflection $T_R : a_\ell^m \mapsto (-1)^m \overline{a_\ell^m}$ is automatically distinguished by odd-parity entries in this block (e.g. $\beta_{2,3,4}$, whose imaginary part flips sign under T_R), so no separate T_R-resolution step is required. For $\ell \geq 5$, the linear bispectral entries at degree ℓ define a system $A_\ell \mathbf{F}_\ell = b_\ell$ with full complex rank $2\ell+1$, and the augmented system (linear bispectral + CG power) has full *real* Jacobian rank, so $\mathbf{F}_\ell$ is uniquely determined. The rank certificates use double-precision arithmetic with well-separated thresholds (residual gaps of $\geq 10^{13}$ relative to machine epsilon) and can be upgraded to interval arithmetic if desired. The full proof is in Appendix A.

3.4 Profiling

We profile the library on a single NVIDIA H100 80 GB GPU (PyTorch 2.10, batch = 16 for the forward pass and 4 for inversion, median wall-clock via `torch.utils.benchmark`); results are in Table 3 and Figure 1.

TODO: Regenerate `experiments/benchmarks/paper_fig1.pdf` to also include D_n-full and disk-full curves once those backends are implemented (currently missing in Figure 1).

TODO: Consider spelling out abbreviated column headers in the table itself (Coefs → Coefficients, Fwd → Forward (ms), Inv → Inversion (ms), Sel. → Selective, Full → Full bispectrum).

The selective forward pass is sub-millisecond for finite abelian modules (C_n: 0.15 ms, $\mathbb{T}^2$: 0.08 ms), under 2 ms for SO(2)/disk (1.94 ms), and under 1 ms for the octahedral group (0.70 ms), making it negligible relative to backbone computation. SO(3) on S^2 at $\ell_{\max}$=16 takes 26.5 ms, dominated by the spherical harmonic transform (via `torch_harmonics` [Bonev et al., 2023]). Selective coefficient counts confirm the $O(|G|)$ scaling: C_{128} uses 128 coefficients vs. 8,256 full (64× reduction); $\mathbb{T}^2$ at

Table 3: Library profiling summary. Timings are median wall-clock on a single NVIDIA H100 80 GB GPU. "Coefs" = number of scalar coefficients (Sel. = selective, Full = full bispectrum); "Fwd" = forward-pass time in milliseconds; "Inv" = inversion time in milliseconds. "–" = not implemented. For continuous groups, $|G| = \infty$; coefficient counts refer to the band-limited representation at the tested resolution. *`SO3onS2` benchmarked at band-limit $\ell_{\max}$=16 (we use $\ell_{\max}$ and L interchangeably to denote the band-limit; see Section 3.3).

Module	G	$\|G\|$	Coefs		Fwd (ms)		Inv (ms)
			Sel.	Full	Sel.	Full	Sel.
`CnonCn`	C_{128}	128	128	8256	0.15	9.17	8.56
`TorusOnTorus`	$C_{32} \times C_{32}$	1024	1024	524800	0.08	0.71	68.02
`DnonDn`	D_{32}	64	245	–	2.52	–	42.33
`SO2onDisk`	$\mathrm{SO}(2)$	∞	411	–	1.94	–	17.69
`SO3onS2`*	$\mathrm{SO}(3)$	∞	430	–	26.54	–	–
`OctaonOcta`	O	24	172	–	0.70	–	–

Table 4: PCam results at matched $\sim$100K parameters (10% training data, mean $\pm$ std over 3 seeds). All equivariant models share a C_8-equivariant DenseNet backbone.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	$.896 \pm .010$	$.826 \pm .012$
Gated	136K	$.941 \pm .009$	$.870 \pm .004$
Bispectrum (`CnonCn`)	128K	$.941 \pm .004$	$\mathbf{.872 \pm .006}$
Fourier-ELU	110K	$\mathbf{.945 \pm .004}$	$.855 \pm .006$
NormReLU	110K	$.942 \pm .004$	$.858 \pm .007$

$|G|$=1,024 uses 1,024 vs. 524,800 full (512×). Inversion is fast for abelian groups (9–68 ms). GPU throughput scales linearly with batch size, exceeding 10^7 samples/s for C_n and $\mathbb{T}^2$ (Appendix C).

4 Experiments

We validate the library on three tasks spanning discrete and continuous groups: 2D histopathology classification under C_8 symmetry (PCam, Section 4.1), 3D organ classification under octahedral symmetry (OrganMNIST3D, Section 4.2), and spherical digit classification under full $\mathrm{SO}(3)$ (Spherical MNIST, Section 4.3).

4.1 2D histopathology classification under cyclic symmetry

We evaluate five invariant pooling strategies on PatchCamelyon [Veeling et al., 2018] (PCam), 327K histopathology patches with binary metastasis labels. All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone following Veeling et al. [2018]; they differ only in the terminal invariant map: norm pooling, gated pooling, Fourier-ELU, and selective bispectrum via `CnonCn`(n=8). A non-equivariant standard CNN trained with data augmentation drawn uniformly from the symmetry group C_8 ("*augmented CNN*") serves as the non-equivariant baseline. We sweep growth rates to vary capacity (30K–1.6M params) and train on 10% of the PCam training set (an image-level subsample, not a group-orbit subsample, drawn before any group augmentation) with AdamW [Loshchilov and Hutter, 2019]. Full model descriptions are in Appendix D.

Results. Table 4 and Figure 2 present the results at matched $\sim$100K parameters.

TODO: Regenerate `experiments/pcam/pareto_analysis.pdf` with body-text-matched fonts, add an "Ours" annotation to the bispectrum bar/curve, and rename panel (c) to "AUC vs. data fraction ($\sim$100K params)" so its title pattern matches panel (a).

At matched capacity ($\sim$100K params, 10% data), all four equivariant pooling methods substantially outperform the standard augmented CNN (AUC 0.896). Fourier-ELU leads slightly (0.945), followed by NormReLU (0.942), bispectrum and gated (both 0.941). At 1% data, the bispectrum

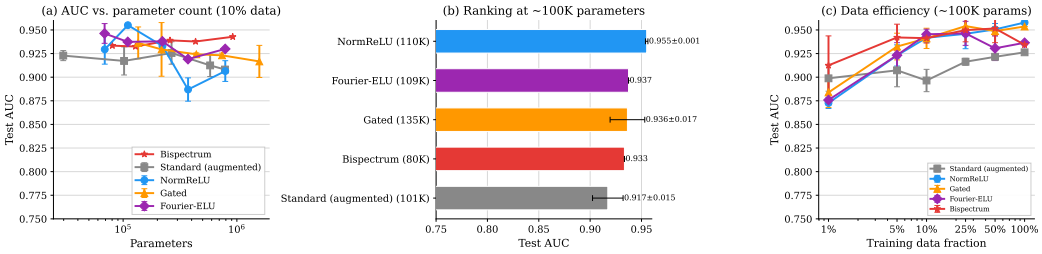

Figure 2: PCam Pareto analysis (single seed, 10% training data). (a) Test AUC vs. parameter count across all five pooling variants. (b) Ranking at the ∼100K parameter budget. (c) Data efficiency at matched ∼100K parameters. Multi-seed statistics are in Table 4.

Table 5: OrganMNIST3D results (mean ± std over 3 seeds). Rot ACC and σ_{rot}: mean and std of accuracy across 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005†

†Nonzero σ_{rot} due to floating-point precision in CG contractions, not a theoretical invariance failure.

leads all methods (0.912), outperforming NormReLU (0.873) and Fourier-ELU (0.876). This data-efficiency advantage narrows as the training set grows, consistent with complete invariants providing a stronger inductive bias when data is scarce.

4.2 3D organ classification under octahedral symmetry

To evaluate the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$), we design a controlled 3D experiment on OrganMNIST3D [Yang et al., 2023] (1,742 abdominal CT volumes, 28^3 voxels, 11 classes). All equivariant variants share an O-equivariant 3D ResNet backbone with exact voxel-permutation kernels. The four models compared are: a translation-equivariant standard 3D CNN trained with octahedral data augmentation (the "*augmented CNN*" baseline, 16K params); the same O-equivariant ResNet with terminal norm pool (375K params); with terminal max pool (374K params); and with terminal selective bispectrum via `OctaonOcta` (463K, 172 invariant scalars). Models are trained with AdamW [Loshchilov and Hutter, 2019] for 100 epochs with three seeds; rotation robustness is measured over all 24 octahedral rotations of the test set. Published baselines and wider-channel experiments are in Appendix E.

Results. Table 5 and Figure 3 report the key findings.

Rotation invariance. All equivariant models achieve $\sigma_{\text{rot}} \leq 0.0005$, while the augmented CNN drops ∼2.5 pp (percentage points) under rotation.

Completeness vs. lossy pooling. The bispectrum (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%) and Regular SE(3) convolutions [Kuipers and Bekkers, 2023] (69.8%). At reduced data (Figure 3), the bispectrum's advantage widens: at 5% data it reaches 33.2% vs. 18.9% for max pool. At higher capacity (6M params, Appendix E), max pool overtakes the bispectrum, suggesting the completeness advantage is specific to the data-limited regime.

TODO: Regenerate `experiments/organ3d/data_efficiency.pdf` to mirror Figure 2 (PCam): add a panel (a) "ACC vs. params" and a panel (b) "Ranking at matched ∼100K params"; draw a chance line at $1/11 \approx 9.1\%$; and either include the norm-pool curve (currently missing) or drop the "norm pool" mention from the prose to match the figure.

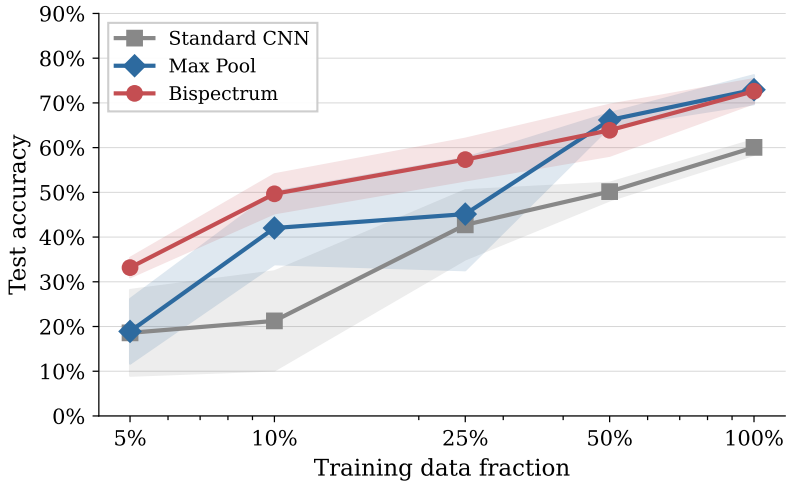

Figure 3: Test accuracy vs. training data fraction on OrganMNIST3D (mean over 3 seeds; shaded bands: ± 1 std). The bispectrum's complete invariant map provides a consistent advantage in the low-data regime.

4.3 Spherical MNIST: SO(3) invariance on the sphere

The preceding experiments validate bispectral pooling for finite groups (C_8 and the octahedral group O). To evaluate the library's continuous group support, we test `SO3onS2`—the $\mathrm{SO}(3)$ bispectrum on the 2-sphere—on Spherical MNIST [Cohen et al., 2018], the standard benchmark for rotation invariance on S^2.

Setup. MNIST digits are projected onto the sphere via stereographic projection and sampled on a 64×128 equiangular grid. We compute the selective bispectrum at $\ell_{\max} = 15$, producing 384 complex invariant coefficients (768 real features after taking log-magnitude), which are fed to a 3-layer MLP (232K parameters total). The completeness proof (Appendix A) covers $4 \leq L \leq 100$, so $\ell_{\max} = 15$ is within the verified range. Three baselines provide controls:

- **Power Spectrum + MLP** (11K params): computes only the rotation-invariant power spectrum $\|f_\ell\|^2$ per degree ℓ, discarding all phase coupling between harmonics. Invariant but *incomplete*.
- **Power Spectrum + MLP (matched)** (166K params): same incomplete invariant features but with a larger MLP (166K), isolating the effect of feature completeness from model capacity.
- **Standard CNN** (185K params): a conventional CNN operating directly on the equirectangular image. Not rotation-invariant.

Following Cohen et al. [2018], we evaluate under three protocols: train on non-rotated / test on non-rotated (NR/NR), train on rotated / test on rotated (R/R), and the most challenging train on non-rotated / test on rotated (NR/R). Each configuration is run with three seeds. Rotation robustness is assessed by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and measuring accuracy variance.

Results. Table 6 and Figure 4 report the results.

Numerically exact $\mathrm{SO}(3)$ invariance. The bispectrum achieves near-identical accuracy across all three protocols ($\sigma_{\mathrm{rot}} \sim 10^{-4}$), confirming true $\mathrm{SO}(3)$ invariance. The standard CNN collapses from 99.1% (NR/NR) to 13.8% (NR/R), showing that augmentation alone does not confer invariance.

Completeness gap. The power spectrum is equally invariant but reaches only 78%—a 17pp deficit that persists even after scaling the MLP to 166K parameters (+1.4pp), confirming incompleteness rather than capacity as the bottleneck.

Table 6: Spherical MNIST results (mean ± std over 3 seeds). Reference numbers are from Cohen et al. [2018]. σ_{rot}: standard deviation of accuracy under 10 random SO(3) rotations (NR-trained models).

Model	Params	NR/NR ↑	R/R ↑	NR/R ↑	σ_{rot} ↓
Cohen planar CNN	—	0.98	0.23	0.11	—
Cohen spherical CNN	—	0.96	0.95	0.94	—
Standard CNN	185K	0.991±0.000	0.781±0.003	0.138±0.003	0.044
Power Spec. + MLP	11K	0.780±0.002	0.780±0.002	0.777±0.001	3×10^{-4}
Power Spec. + MLP (matched)	166K	0.794±0.011	0.794±0.011	0.792±0.010	3×10^{-4}
Bispectrum + MLP (ours)	232K	**0.950±0.000**	**0.949±0.002**	**0.951±0.001**	$\mathbf{1\times10^{-4}}$

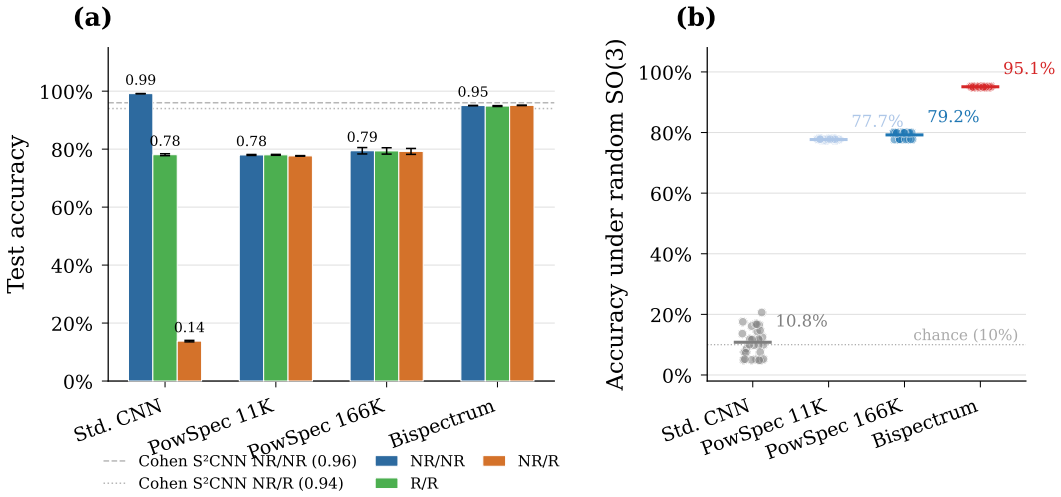

Figure 4: Spherical MNIST analysis. **(a)** Cohen et al. protocol: NR/NR, R/R, and NR/R accuracy. Dashed lines show Cohen's spherical CNN reference. **(b)** Accuracy under 10 random SO(3) rotations (NR-trained models, 30 points per model = 10 rotations × 3 seeds). Scaling the power spectrum MLP from 11K to 166K parameters yields only +1.4pp; the bispectrum at 232K reaches 95.0%.

Comparison with Cohen et al. Cohen's spherical CNN [Cohen et al., 2018] reaches 96%/95%/94% with learned equivariant convolutions. Our single-pass bispectrum (95.0%/94.9%/95.1%) is 1–2pp lower but has a tighter invariance gap (0.1pp vs. 2.0pp).

5 Related Work

Equivariant network libraries. `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022] provide general-purpose equivariant convolution layers for 2D and 3D groups respectively, and `lie-nn` [Batatia et al., 2023] provides general Lie-group representation primitives (irreducible representations, Clebsch–Gordan decompositions, tensor products) on top of which custom equivariant layers can be built. These libraries implement equivariant *feature maps* but leave the choice of invariant pooling to the user (typically norm or gate). Our library is complementary: it provides the invariant map as a drop-in `nn.Module` that plugs into architectures built with any of these frameworks.

Bispectral approaches in neural networks. Sanborn et al. [2023] introduced bispectral neural networks (BNNs), using the full G-bispectrum as an invariant nonlinearity for C_n and D_n. Oreiller et al. [2022] applied bispectral CNNs to medical imaging with a fixed set of hand-selected triples. Both use the full (quadratic-cost) bispectrum and provide implementations tied to specific architectures. Our library generalises to seven group/domain pairs, defaults to the $O(|G|)$ selective reduc-

tion, and exposes a framework-agnostic API. Chevalley et al. [2024] explore bispectral signatures for data characterisation, using bispectra as descriptors rather than learnable pooling primitives. Outside the equivariant-network setting, `PyBispectra` [Binns et al., 2025] computes bispectrum and bicoherence statistics of one-dimensional time-series signals for electrophysiology applications such as phase-amplitude coupling and time-delay estimation; the underlying 1D bispectrum is translation-invariant by construction, but the toolbox is not designed as a neural-network pooling layer.

CG product nonlinearities and tensor products. Tensor field networks [Thomas et al., 2018] and MACE [Batatia et al., 2022] use iterated Clebsch–Gordan products to build equivariant interactions. These products are exact and complete in principle, but each layer incurs $O(L^3)$ cost per CG contraction. Xie et al. [2025] analyse the *price of freedom* in tensor product architectures, showing that maintaining full expressivity through CG products becomes expensive at high angular momenta. The augmented selective bispectrum achieves completeness in a single $O(L^2)$ pass (for $\mathrm{SO}(3)$) or $O(|G|)$ pass (for finite groups), avoiding iterated contractions.

Invariant theory and scattering. The wavelet scattering transform [Mallat, 2012] provides translation-invariant representations via iterated modulus operations; Esteves et al. [2018] extend this to the sphere. Unlike the bispectrum, scattering representations are stable but generally incomplete—they discard inter-scale phase information. The bispectrum's completeness guarantee [Kakarala, 2012] is its distinguishing theoretical advantage.

6 Discussion and Conclusion

We have presented `bispectrum`, a PyTorch library that makes selective G-bispectral invariants practical for machine learning. The library covers seven group/domain pairs with a uniform `nn.Module` interface and required one technical contribution during implementation: an augmented selective $\mathrm{SO}(3)$ bispectrum on S^2 achieving the optimal $O(L^2)$ coefficient count (Theorem 1). Table 3 confirms that the selective forward pass adds negligible overhead to training pipelines ($<$1 ms for all finite-group modules on a single GPU). Across three experimental settings, bispectral pooling provides numerically exact rotation invariance and a data-efficiency advantage over incomplete alternatives in the low-data / moderate-capacity regime, where the complete invariant map provides a stronger inductive bias that partially compensates for limited training data.

When not to use the bispectrum. The data-efficiency advantage of complete invariants is most pronounced in the low-data, moderate-capacity regime. At high model capacity (Section 4), incomplete alternatives such as max pooling can match or surpass bispectral pooling, likely because the network learns to compensate for missing invariant information. Completeness is a property of the *map*, not of downstream task performance: a complete invariant provides a stronger inductive bias, but does not guarantee superior accuracy on every task. For $\mathrm{SO}(3)$ on S^2, computational cost grows as $O(L^2)$ per forward pass; at very large band-limits this may become a bottleneck relative to simpler pooling strategies.

Limitations. Inversion for $\mathrm{SO}(3)$ on S^2 is not yet implemented; only the selective forward map is available. The selective $\mathrm{SO}(3)$ completeness result is a *computer-assisted* theorem: it relies on a genericity assumption (the signal must avoid a measure-zero algebraic variety) and on per-degree rank certificates discharged by deterministic numerical witnesses (seed 42) with $\sigma_{\min}/\sigma_{\max} \geq 10^{-4}$ in double precision (margin $>10^{11}$ above double-precision rounding error) for $4 \leq L \leq 100$; the constant-size seed certificate at $L_0 = 4$ (Lemma 39) is verified in exact rational arithmetic via `sympy`. A structural, all-L proof (i.e. a uniform-in-ℓ rank argument) and a full characterisation of the degenerate set remain open. Our experimental comparisons do not include tensor-product architectures (e3nn, MACE) or wavelet scattering, which target different input assumptions and would require non-trivial adaptation to the benchmarks used here.

Future work. Key directions include: implementing $\mathrm{SO}(3)$ inversion from selective bispectral coefficients; a structural proof of bootstrap rank for all L; numerical conditioning analysis at large band-limits; integration with `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022]; and applications to cryo-EM reconstruction and molecular property prediction.

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A Selective $\mathrm{SO}(3)$ Bispectrum – Full Proof

This appendix provides the complete proof of Theorem 1 from the main text, establishing completeness of the augmented selective $\mathrm{SO}(3)$-bispectrum on S^2 for band-limits $4 \leq L \leq 100$ (extendable to any desired L by re-running the verification script).

Table 7: Per-degree *bootstrap* entry counts in the augmented selective $\mathrm{SO}(3)$ invariant, **excluding the constant-size full $L_0 = 4$ seed block** added by the seed step of Algorithm 1. The "Bispec" column counts scalar bispectral entries including mandatory even self-coupling entries $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \leq \ell' \leq \ell$); entries that vanish for real signals are excluded. The "CG Power" column counts CG power spectrum augmentations. At $\ell = 1$, $\beta_{1,1,0}$ is excluded (redundant with power). At $\ell = 2$, $\beta_{1,2,1}$ is excluded (proportional to $\beta_{1,1,2}$ after gauge-fixing).

ℓ	Bispec	CG Power	Total
0	1	0	1
1	1	0	1
2	3	1	4
3	7	4	11
4	12	5	17
$\ell \geq 5$	$\sim 2\ell{+}3$	$\sim \ell$	$\sim 3\ell$

The theorem-level invariant additionally includes a full-bispectrum seed block at $L_0 = 4$ contributing at most 30 scalar entries (independent of L); the total invariant size therefore remains $\Theta(L^2)$.

A.1 Setup and notation

Definition 3 (Band-limited signal on S^2)**.** A signal $f : S^2 \to \mathbb{R}$ is *band-limited* at degree L if its spherical harmonic expansion

$$f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m \, Y_\ell^m(\theta, \varphi) \tag{3}$$

has $a_\ell^m = 0$ for all $\ell > L$. We write $\mathbf{F}_\ell = (a_\ell^{-\ell}, \ldots, a_\ell^{\ell}) \in \mathbb{C}^{2\ell+1}$ for the coefficient vector at degree ℓ.

For real f, the conjugation symmetry $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ holds, so $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The total signal dimension is $\dim = \sum_{\ell=0}^{L}(2\ell + 1) = (L + 1)^2$.

Definition 4 (Rotation action)**.** Under $g \in \mathrm{SO}(3)$, the spherical harmonic coefficients transform as $\mathbf{F}_\ell \mapsto D^\ell(g)\,\mathbf{F}_\ell$, where D^ℓ is the $(2\ell+1)$-dimensional Wigner D-matrix.

Definition 5 (Bispectrum on S^2)**.** For each triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$, the *bispectral coefficient* is

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,\, m_2,\, m \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}\; a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2}\, \overline{a^{m}_{\ell}}, \tag{4}$$

where $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$ denotes the Clebsch–Gordan coefficient $\langle \ell_1, m_1;\, \ell_2, m_2 \mid \ell, m\rangle$. Each $\beta_{\ell_1,\ell_2,\ell}$ is a scalar; for real f it is real when $\ell_1+\ell_2+\ell$ is even and purely imaginary when $\ell_1+\ell_2+\ell$ is odd (Lemma 27).

Definition 5 is the specialization of the general G-bispectrum (eq. (2)) to the case $G = \mathrm{SO}(3)$ acting on S^2, where the irreps are Wigner-D matrices and the Fourier coefficients are spherical harmonic coefficients a^m_ℓ.

Definition 6 (CG power spectrum)**.** For a triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality, the *CG power spectrum entry* is

$$P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2 = \sum_{m=-\ell}^{\ell} \Bigg| \sum_{\substack{m_1,m_2 \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}\; a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2} \Bigg|^2. \tag{5}$$

This is a degree-4 polynomial in the signal coefficients (real and non-negative for real f) and $\mathrm{SO}(3)$-invariant: the CG-coupled product transforms as $D^\ell(g)$, and $\|\cdot\|^2$ is invariant under unitary transformations.

Proposition 7 (Size of the full bispectrum)**.** *The number of admissible triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1, \ell_2, \ell \le L$ satisfying the triangle inequality $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$ is $\Theta(L^3)$.*

Proof. For fixed ℓ_1 and ℓ_2, the triangle inequality admits $\ell \in \{|\ell_1 - \ell_2|, \ldots, \min(\ell_1 + \ell_2, L)\}$, giving $\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1$ valid values of ℓ. Summing:

$$N_{\text{full}} = \sum_{\ell_1=0}^{L} \sum_{\ell_2=0}^{L} \big(\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1\big) = \tfrac{1}{3}L^3 + \mathcal{O}(L^2).$$

□

Remark 8 (Invariance). Each $\beta_{\ell_1,\ell_2,\ell}$ is $\mathrm{SO}(3)$-invariant: the Wigner matrices satisfy $D^{\ell_1} \otimes D^{\ell_2} = \bigoplus_\ell D^\ell$ under the CG decomposition, so the triple product contracted with CG coefficients is unchanged by $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$.

A.2 Why $\mathcal{O}(L)$ is impossible

Proposition 9 (Dimensional lower bound)**.** *Any complete $\mathrm{SO}(3)$-invariant for band-limited signals on S^2 at band-limit L must have at least $(L+1)^2 - 3$ functionally independent components.*

Proof. The orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L+1)^2 - \dim \mathrm{SO}(3) = (L+1)^2 - 3$. Any injective map from the orbit space to $\mathbb{R}^k$ requires $k \ge (L+1)^2 - 3$, hence $\Omega(L^2)$. □

Remark 10. The chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \ldots, L-1$ provides only $\sim 3L$ scalar constraints (since each chain entry couples $\mathbf{F}_1 \in \mathbb{C}^3$ with one known $\mathbf{F}_\ell$). This cannot recover $(L+1)^2 \approx L^2$ unknowns.

A.3 The selective index set

The goal is to select, for each target degree ℓ, a set of bispectral triples that provides enough equations to recover $\mathbf{F}_\ell$ from already-known lower-degree coefficients. Since $\mathbf{F}_\ell$ has $2\ell+1$ real degrees of freedom, we need at most $2\ell+1$ entries. We classify candidate triples by how $\mathbf{F}_\ell$ enters the bispectral formula (4): *linearly* (when it occupies a position where the other two indices are known), or *nonlinearly* (when it appears more than once).

Definition 11 (Selective bispectrum on S^2)**.** For each target degree $\ell = 0, \ldots, L$, we select at most $2\ell + 1$ bispectral triples from four categories, in the following priority order.

Category 1 (Chain). Triples (ℓ_1, ℓ_2, ℓ) with $\ell_1 \leq \ell_2 < \ell$ and $\ell_1 + \ell_2 \geq \ell$. Linear in $\mathbf{F}_\ell$: the unknown $\overline{a_\ell^m}$ appears once, multiplied by known lower-degree coefficients.

Category 2 (Cross). Triples (ℓ_1, ℓ, ℓ') with $1 \leq \ell_1 < \ell$ and $\ell - \ell_1 \leq \ell' < \ell$. Here $\mathbf{F}_\ell$ appears in the second (tensor-product) position, yielding equations linear in $a_\ell^{m_2}$ with known $a_{\ell_1}^{m_1}$ and $a_{\ell'}^m$.

Category 3 (Power). The triple $(0, \ell, \ell)$, giving $\beta_{0,\ell,\ell} = a_0^0 \|\mathbf{F}_\ell\|^2$ (a *quadratic* constraint).

Category 4 (Self-coupling). Triples (ℓ, ℓ, ℓ') with $0 \leq \ell' < \ell$. Quadratic in $\mathbf{F}_\ell$. Selected only for the remaining deficit at low degrees.

Proposition 12 (Bispectral entry counts)**.** *At $\ell = 4$ the explicit block $\mathcal{T}_4^{\text{small}}$ has 10 linear entries (all chain+cross candidates). For each $5 \leq \ell \leq 7$, the explicit block $\mathcal{T}_\ell^{\text{small}}$ has exactly $2\ell+1$ linear entries. For each $\ell \geq 8$, the proof-first linear family*

$$\mathcal{T}_\ell := \{(a, \ell, \ell-a) : 1 \leq a \leq \ell-1\} \cup \{(a, \ell, \ell-a+1) : 2 \leq a \leq \ell-1\} \cup \{(a, \ell-a, \ell) : 1 \leq a \leq 4\}$$

also has exactly $2\ell + 1$ entries.

Proposition 13 (Augmented output size)**.** *For $L \geq 4$, the bispectral component $\mathcal{S}_\beta$ has $(L+1)^2 - 3$ budget-selected entries plus $\sum_{\ell=4}^{L} \lfloor \ell/2 \rfloor$ mandatory even self-coupling entries, giving $\Theta(L^2)$ bispectral scalars. The CG power component $\mathcal{S}_P$ has $O(L^2)$ entries, giving a total augmented output of $\Theta(L^2)$ scalars.*

Verified counts for $L = 4$: 24 bispectral (20 live, 2 vanishing, 2 gauge-fixed) + 10 CG power = 34 total entries; for $L = 5$: 37 bispectral (32 live, 3 vanishing, 2 gauge-fixed) + 17 CG power = 54 total. The Jacobian-rank certificate (Lemma 35) uses only the $m = 30$ (resp. 49) live entries.

Proof. Bispectral entries. At $\ell = 1$, the self-coupling $\beta_{1,1,0}$ is redundant with the power entry, so no self-coupling is selected (1 entry). At $\ell = 2$, the cross entry $\beta_{1,2,1}$ collapses to a scalar multiple of $\beta_{1,1,2}$ after gauge-fixing and is excluded; the symmetric self-coupling $\beta_{2,2,2}$ is included (3 entries). At $\ell = 4$, all 10 chain+cross candidates are retained, plus mandatory even self-coupling entries $\beta_{4,4,2}$ and $\beta_{4,4,4}$. Similarly, at each $\ell \geq 3$, the even self-coupling entries $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \leq \ell' \leq \ell$) add $\lfloor \ell/2 \rfloor$ entries beyond the linear-block budget. Therefore:

$$|\mathcal{S}_\beta| = (L+1)^2 - 3 + \sum_{\ell=4}^{L} \lfloor \ell/2 \rfloor = \Theta(L^2).$$

CG power entries. At each degree $\ell \geq 2$, the CG power augmentation adds at most $2\ell+1$ entries (one per real unknown in $\mathbf{F}_\ell$). Summing gives an upper bound:

$$|\mathcal{S}_P| \;\leq\; \sum_{\ell=2}^{L} (2\ell+1) \;=\; L^2 + 2L - 3 \;=\; O(L^2).$$

Total. $|\mathcal{S}_\beta| + |\mathcal{S}_P| = (L+1)^2 - 3 + O(L^2) = \Theta(L^2)$. The lower bound $\Omega(L^2)$ follows from $|\mathcal{S}_\beta|$ alone. □

A.4 Completeness proof

The proof proceeds in three stages: (1) recovery of $\mathbf{F}_0$, $\mathbf{F}_1$ and $\mathrm{SO}(3)$ gauge fixing; (2) structural seed completeness for degrees 0–L_0 ($L_0 \geq 2$) via Kakarala's bispectrum completeness theorem applied to the $\Theta(1)$-extra full-bispectrum block at L_0; (3) per-degree linear bootstrap for $\ell > L_0$. The key difference from the finite-group case [Mataigne et al., 2024] is that on S^2, the scalar bispectrum alone has rank deficiency for real signals, necessitating CG power augmentation at every degree.

For the seed (stage 2), we adjoin to Φ_{aug} the full bispectrum at a fixed low cutoff L_0 (we use $L_0 = 4$, contributing $\leq$30 scalar entries, independent of L). Kakarala's theorem then gives structural seed recovery as a single black-box citation; Appendix B contains an *explicit constructive solver* for the same seed step but is not required for the completeness statement.

Algorithm 1 Augmented selective index set for the $\mathrm{SO}(3)$ bispectrum on S^2 at band-limit L. Returns bispectral triples $\mathcal{S}_\beta$ and CG power triples $\mathcal{S}_P$.

Input: Band-limit L
Output: Bispectral set $\mathcal{S}_\beta$, CG power set $\mathcal{S}_P$
$\mathcal{S}_\beta \leftarrow \{(0,0,0)\}$; $\mathcal{S}_P \leftarrow \emptyset$
Seed block: add to $\mathcal{S}_\beta$ all triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1 \le \ell_2 \le L_0 = 4$ and $|\ell_1 - \ell_2| \le \ell \le \min(\ell_1 + \ell_2, L_0)$ {constant-size full-bispectrum seed required by Lemma 24}
for $\ell = 1, \ldots, L$ **do**
  budget $\leftarrow 2\ell + 1$
  if $\ell = 4$ **then**
    budget $\leftarrow 10$ {keep all chain+cross at $\ell = 4$}
  end if
  candidates $\leftarrow [\,]$
  if $4 \le \ell \le 7$ **then**
    append the explicit low-degree linear block $\mathcal{T}_\ell^{\mathrm{small}}$ to candidates
  else if $\ell \ge 8$ **then**
    for $a = 1, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a)$ to candidates
    end for
    for $a = 2, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a{+}1)$ to candidates
    end for
    for $a = 1, 2, 3, 4$ **do**
      append $(a,\ \ell{-}a,\ \ell)$ to candidates
    end for
  else
    append low-degree chain candidates
    if $\ell \ne 2$ **then**
      append cross candidates {$\beta_{1,2,1}$ excluded}
    end if
  end if
  append $(0,\ \ell,\ \ell)$ to candidates {power entry}
  if $\ell \notin \{1,\ 2\}$ **then**
    for $\ell' = \ell{-}1, \ldots, 0$ **do**
      append $(\ell,\ \ell,\ \ell')$ to candidates
    end for
  end if
  if $\ell = 2$ **then**
    append $(\ell,\ \ell,\ \ell)$ to candidates {$\beta_{2,2,2}$}
  end if
  Remove entries that vanish for real signals; deduplicate against $\mathcal{S}_\beta$ already added by the seed block
  $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup$ first $\min(\text{budget},\ |\text{candidates}|)$ live entries
  {**Mandatory even self-couplings**}
  for $\ell' = 2, 4, \ldots, \lfloor \ell/2 \rfloor \cdot 2$ with $\ell' \le \ell$ and $\ell \ge 3$ **do**
    $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup \{(\ell,\ \ell,\ \ell')\}$
  end for
  {**CG power augmentation**}
  Greedily add $P_{\ell_1,\ell_2,\ell'}$ (with $\max(\ell_1, \ell_2) = \ell$, $|\ell_1{-}\ell_2| \le \ell' \le \ell_1{+}\ell_2$) to $\mathcal{S}_P$ until the Jacobian rank of Φ_{aug} at degree ℓ matches the number of unknowns
end for
return $(\mathcal{S}_\beta,\ \mathcal{S}_P)$

Definition 14 (Reflection map T_R)**.** The azimuthal reflection $R : (\theta, \varphi) \mapsto (\theta, -\varphi)$ acts on spherical harmonic coefficients as $a_\ell^m \mapsto (-1)^m a_\ell^{-m}$. For real signals, this equals complex conjugation: $a_\ell^m \mapsto \overline{a_\ell^m}$. Since $R \in \mathrm{O}(3) \setminus \mathrm{SO}(3)$, the signal $R \cdot f$ is generically in a *different* $\mathrm{SO}(3)$-orbit from f.

Assumption 15. The signal f satisfies:

(G1) $a_0^0 \ne 0$ (non-zero mean).
(G2) $\|\mathbf{F}_1\| \ne 0$ (non-zero dipole).
(G3) After gauge-fixing (Lemma 18), $a_2^1 \ne 0$ (azimuthal gauge is resolvable).

Remark 16. Each condition above excludes a proper algebraic subvariety of the signal space $\mathbb{R}^{(L+1)^2}$; their union is Lebesgue-null. The proof uses additional genericity conditions—seed Jacobian rank, bootstrap rank, and degree-4 consistency—but each is *proved* to hold outside a proper algebraic subvariety (not assumed).

Theorem 17 (Augmented selective $\mathrm{SO}(3)$ invariant on S^2: computer-assisted completeness for $4 \le L \le 100$)**.** *For $4 \le L \le 100$, let $f, f' : S^2 \to \mathbb{R}$ be band-limited at degree L and satisfy Assumption 15. If $\Phi_{\mathrm{aug}}(f) = \Phi_{\mathrm{aug}}(f')$ (as complex-valued output, including both real and imaginary parts of each bispectral entry), then $f' = g \cdot f$ for some $g \in \mathrm{SO}(3)$.*

The generic set (satisfying (G1)–(G3) and lying outside a proper algebraic subvariety) has full Lebesgue measure in V_L. The augmented output has $\Theta(L^2)$ real components, achieving the information-theoretic lower bound up to a constant factor.

Lemma 18 (Gauge fixing)**.** *For generic $f \in V_L$ (with $\|\mathbf{F}_1\| \neq 0$ and $a_2^1 \neq 0$ after aligning $\mathbf{F}_1$), there exists a unique $g \in \mathrm{SO}(3)$ such that the gauge-fixed signal $\tilde{f} = g \cdot f$ satisfies:*

(i) $a_1^{-1} = a_1^1 = 0, \quad a_1^0 = c > 0$ *(alignment of $\mathbf{F}_1$ to the z-axis),*
(ii) $\mathrm{Im}(a_2^1) = 0, \quad a_2^1 = x > 0$ *(fixing the residual $\mathrm{SO}(2)_z$ symmetry).*

This consumes all 3 degrees of freedom of $\mathrm{SO}(3)$.

Proof. The $\ell = 1$ Wigner D-matrix $D^1(g)$ is the standard 3×3 rotation matrix. Since $\mathrm{SO}(3)$ acts transitively on S^2, any nonzero $\mathbf{F}_1$ can be rotated to align with $\hat{z}$, leaving an $\mathrm{SO}(2)_z$ stabiliser. Choosing $\varphi = -\arg(a_2^1)$ yields $a_2^1 \in \mathbb{R}_{>0}$, exhausting all three rotational degrees of freedom. □

Remark 19 (Gauge-fixed parameterisation). On the gauge slice, the real parameters are:

$$A_0 = a_0^0, \quad c = a_1^0, \quad (y, x, u, v) = (a_2^0, \mathrm{Re}\, a_2^1, \mathrm{Re}\, a_2^2, \mathrm{Im}\, a_2^2),$$

and for $\ell \ge 3$: $t_\ell = a_\ell^0$, and $(p_{\ell,m}, q_{\ell,m}) = (\mathrm{Re}\, a_\ell^m, \mathrm{Im}\, a_\ell^m)$ for $m = 1, \ldots, \ell$. The reflection T_R (Definition 14) acts on the gauge slice as:

$$T_R : \quad v \mapsto -v, \qquad q_{\ell,m} \mapsto -q_{\ell,m} \quad \forall\, \ell \ge 3,\ m \ge 1. \tag{6}$$

All other gauge-fixed parameters are unchanged. T_R preserves the gauge conditions (i)–(ii).

Lemma 20. *For $a_0^0 \neq 0$: $\beta_{0,0,0} = (a_0^0)^3$ determines $A_0 = a_0^0$ uniquely. Then $\beta_{0,1,1} = A_0 \|\mathbf{F}_1\|^2$ determines $c = \|\mathbf{F}_1\|$, and $\mathbf{F}_1 = (0, c, 0)^\top$ is fixed by the gauge.*

A.4.1 Structural seed completeness via Kakarala's theorem

Lemma 21 (Kakarala bispectrum completeness on S^2, restated)**.** *Let $G = \mathrm{SO}(3)$ and $H = \mathrm{SO}(2)_z \subset G$ be the stabiliser of the north pole, so that $G/H \cong S^2$. Let $r, s \in L^1(G)$ be left H-invariant functions whose Fourier coefficients $R(\ell), S(\ell)$ all satisfy the maximal H-rank condition (equivalently, for S^2 functions, the spherical-harmonic vectors $\mathbf{F}_\ell$ are nonzero for every ℓ). Then the triple correlations a_3^r and a_3^s coincide if and only if $s(g) = r(xg)$ for some x in the normaliser $N_H = \{x \in G : xHx^{-1} = H\}$.*

Proof. This is [Kakarala, 2012, Theorem 7], originally proved in [Kakarala, 1992]; the proof proceeds via the Iwahori–Sugiura duality theorem for homogeneous spaces of compact groups. An independent computer-algebra certificate of the same conclusion for $L \le 15$ (Gröbner-basis elimination, no numerical threshold) is given by Bandeira et al. [2017], which we invoke below as a backup whenever a structural reader prefers to avoid relying on the thesis [Kakarala, 1992] for the proof of Theorem 7. □

Remark 22 (Computation of N_H and absence of an $\mathrm{O}(3)$ ambiguity). For $H = \mathrm{SO}(2)_z = \{R_z(\varphi) : \varphi \in \mathbb{R}\}$, an element $x \in \mathrm{SO}(3)$ lies in N_H iff $x \cdot \hat{z} = \pm\hat{z}$, hence $N_H = H \cup R_x(\pi) \cdot H$, the dihedral group $\mathrm{O}(2)$ realised *inside* $\mathrm{SO}(3)$. In particular, $N_H \subset \mathrm{SO}(3)$: the non-trivial coset representative $R_x(\pi)$ is a proper $\mathrm{SO}(3)$-rotation (an azimuthal-axis flip), *not* an $\mathrm{O}(3) \setminus \mathrm{SO}(3)$ reflection. Therefore Lemma 21 immediately implies $\mathrm{SO}(3)$-orbit recovery on S^2: equality of bispectra forces s and r to lie in the same $\mathrm{SO}(3)$-orbit, and the residual $N_H/H = \mathbb{Z}/2$ is internal to $\mathrm{SO}(3)$ (it merely reflects that two distinct H-invariant lifts of the same S^2-function exist along an $\mathrm{SO}(3)$-orbit). The orientation-reversing reflection $T_R : a_\ell^m \mapsto (-1)^m \overline{a_\ell^m}$ (Definition 14) is a separate operation: $T_R \in \mathrm{O}(3) \setminus \mathrm{SO}(3)$ flips the sign of every odd-parity bispectral entry (Lemma 27), so $T_R \cdot f$ is generically detected as a different signal by the full bispectrum.

Definition 23 (Augmented invariant with L_0-seed block)**.** Fix a seed cutoff L_0 with $4 \le L_0 \le L$. We augment Φ_{aug} from Definition 11 with the *full* bispectrum at L_0, i.e. all triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1 \le \ell_2 \le L_0$ and $|\ell_1 - \ell_2| \le \ell \le \min(\ell_1 + \ell_2, L_0)$. The number of additional entries is $N_{\mathrm{full}}(L_0) = \frac{1}{3}L_0^3 + O(L_0^2)$, a *constant* independent of L (for $L_0 = 4$ at most 30 scalars). The output size of the augmented invariant is therefore

$$|\Phi_{\mathrm{aug}}^{\mathrm{seed}+}| = \Theta(L^2) + \Theta(L_0^3) = \Theta(L^2),$$

which preserves the asymptotic budget.

Lemma 24 (Structural seed recovery)**.** *Fix L_0 with $4 \le L_0 \le 15$ (we use $L_0 = 4$). For every signal $f \in V_L$ satisfying Assumption 15 and the open genericity condition $\mathbf{F}_\ell \neq 0$ for $0 \le \ell \le L_0$, the full bispectrum at degree L_0 (adjoined to Φ_{aug} as in Definition 23) uniquely determines $(\mathbf{F}_0, \mathbf{F}_1, \ldots, \mathbf{F}_{L_0})$ on the gauge slice of Lemma 18.*

Proof. View the band-limit-L_0 truncation of f as an H-invariant function on $G = \mathrm{SO}(3)$ (with $H = \mathrm{SO}(2)_z$): the genericity condition $\mathbf{F}_\ell \neq 0$ is exactly the maximal H-rank hypothesis of Lemma 21. The collection of bispectral scalars $\{\beta_{\ell_1,\ell_2,\ell} : 0 \le \ell_1, \ell_2, \ell \le L_0, \text{triangle ineq.}\}$ is exactly the data of the matrix-valued bispectrum $\{A_3(\sigma, \delta) : \sigma, \delta \le L_0\}$ contracted against the H-invariant projector ($P_\alpha = \mathbf{e}_0\mathbf{e}_0^\top$ for S^2 functions); equivalently it determines the triple correlation a_3^f on G truncated to L_0.

Two independent arguments establish the conclusion:

(i) *(Representation-theoretic.)* By Lemma 21 and Remark 22, two generic L_0-truncated signals with the same full bispectrum at L_0 are N_H-translates with $N_H \subset \mathrm{SO}(3)$, hence lie in the same $\mathrm{SO}(3)$-orbit.

(ii) *(Computer-algebra.)* Bandeira et al. [2017] verify by symbolic computation (Jacobian rank of the bispectrum map at a rational test point, combined with Zariski-density propagation) that the bispectrum separates generic $\mathrm{SO}(3)$-orbits on $V_0 \oplus \cdots \oplus V_{L_0}$ for every $L_0 \le 15$. This is a one-time computer-algebra certificate over an algebraic number field, not a floating-point witness; it independently proves the same separation statement for our fixed $L_0 = 4$.

Either argument suffices. The gauge slice of Lemma 18 fixes a representative in the resulting $\mathrm{SO}(3)$-orbit, so the gauge-fixed seed coordinates $(\mathbf{F}_0, \ldots, \mathbf{F}_{L_0})$ are uniquely determined. □

Remark 25 (On the "up to reflection" caveat of Edidin and Satriano [2024]). Edidin and Satriano [2024, Section 1] write that Kakarala [2012] "recovers the $\mathrm{SO}(3)$-orbit of a band-limited real-valued function up to reflection from the bispectrum." This caveat is sharp for the *constructive* algorithm in Kakarala's Section 5 (where the positive-definite square root of $\hat{F}(1)\hat{F}(1)^\dagger$ pins $\hat{F}(1)$ only up to right multiplication by a real orthogonal matrix, an $\mathrm{O}(3)$-ambiguity that the recursion then propagates), but it is *not* present in Theorem 7 [Kakarala, 2012] as stated, which gives recovery up to $N_H \subset \mathrm{SO}(3)$. The appearance of an $\mathrm{O}(3) \setminus \mathrm{SO}(3)$ ambiguity is therefore an artefact of the recursion's polar-decomposition step, not of the bispectrum data itself: the odd-parity entries $\beta_{\ell_1,\ell_2,\ell}$ with $\ell_1 + \ell_2 + \ell$ odd (the smallest of which is $\beta_{2,3,4}$, all-distinct indices) flip sign under the orientation-reversing reflection $T_R : a_\ell^m \mapsto (-1)^m \overline{a_\ell^m}$ (Lemma 27), so the full bispectrum at any $L_0 \ge 4$ already separates f from $T_R \cdot f$. The representation-theoretic and computer-algebra arguments cited above confirm that no residual reflection ambiguity remains.

Remark 26 (Relation to the constructive seed lemmas). Appendix B provides an *explicit constructive solver* for the seed: closed-form recovery at degree 2, finite-fibre enumeration at degree 3, degree-4 filtering, and an explicit parity-based T_R resolution. This algorithmic content is used by the implementation but is no longer required for the completeness statement, which follows directly from Lemma 24. The constructive solver remains useful because it reduces the seed to a low-dimensional, well-conditioned numerical solve, whereas the Kakarala route leaves the seed inversion implicit (the proof is existential).

A.4.2 Parity structure of the bispectrum

Lemma 27 (Parity of the bispectrum)**.** *For any signal (real or complex):*

$$\beta_{\ell_1,\ell_2,\ell}(T_R \cdot f) = (-1)^{\ell_1+\ell_2+\ell}\, \beta_{\ell_1,\ell_2,\ell}(f). \tag{7}$$

For real *signals, there is an independent constraint:*

$$\overline{\beta_{\ell_1,\ell_2,\ell}(f)} = (-1)^{\ell_1+\ell_2+\ell}\,\beta_{\ell_1,\ell_2,\ell}(f). \tag{8}$$

Together:

(a) If $\ell_1+\ell_2+\ell$ is even: β is real-valued (by (8)*) and T_R-invariant (by* (7)*).*
(b) If $\ell_1+\ell_2+\ell$ is odd: β is purely imaginary (by (8)*) and $\beta(T_R\cdot f) = -\beta(f)$, so $\operatorname{Im}\beta$ flips sign under T_R.*

Proof. Under T_R, each $a_\ell^m \mapsto (-1)^m a_\ell^{-m}$. Substituting into $\beta = \sum_m(\sum_{m_1+m_2=m} C\, a^{m_1}a^{m_2})\overline{a^m}$, the three sign factors $(-1)^{m_1+m_2+m} = (-1)^{2m} = 1$ cancel. Re-indexing $m \mapsto -m$ and applying the CG symmetry $C^{\ell,\,-m}_{\ell_1,\,-m_1;\,\ell_2,\,-m_2} = (-1)^{\ell_1+\ell_2-\ell}\, C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}$ yields (7) (with $(-1)^{\ell_1+\ell_2-\ell} = (-1)^{\ell_1+\ell_2+\ell}$).

For real signals, $\overline{a_\ell^m} = (-1)^m a_\ell^{-m}$. Taking the complex conjugate of β and re-indexing similarly yields (8). □

Lemma 28 (CG power is T_R-invariant)**.** *For all triples (ℓ_1,ℓ_2,ℓ): $P_{\ell_1,\ell_2,\ell}(T_R\cdot f) = P_{\ell_1,\ell_2,\ell}(f)$.*

Proof. Under T_R, the CG-projected component transforms as $(\mathbf{F}_{\ell_1}\otimes\mathbf{F}_{\ell_2})|_\ell^M \mapsto (-1)^{\ell_1+\ell_2+\ell+M}\,(\mathbf{F}_{\ell_1}\otimes\mathbf{F}_{\ell_2})|_\ell^{-M}$. Taking the squared norm and re-indexing the sum over M leaves P unchanged. □

Proposition 29 (Vanishing of odd-parity entries at low degrees)**.** *For real signals, the odd-parity bispectral entry $\beta_{\ell_1,\ell_2,\ell}$ (with $\ell_1+\ell_2+\ell$ odd) vanishes identically whenever two of the three representation indices coincide. In particular:*

(i) $\ell_1=\ell_2$: the CG coefficients are antisymmetric under swapping $m_1\leftrightarrow m_2$, while $a_{\ell_1}^{m_1}a_{\ell_1}^{m_2}$ is symmetric, giving exact cancellation.
(ii) Two of $\{\ell_1,\ell_2,\ell\}$ are equal: a symmetry argument yields $\beta = -\overline{\beta}$ ***and*** *$\beta = 0$, verified symbolically for all triples with indices ≤ 5.*

Corollary 30 (First nonzero odd-parity entry)**.** *The first bispectral entry with odd parity and generically nonzero imaginary part occurs at degree $\ell = 4$: $\beta_{2,3,4}$, with $\ell_1+\ell_2+\ell = 9$ (odd) and ℓ_1,ℓ_2,ℓ all distinct. At the deterministic witness (seed 42), $\operatorname{Im}(\beta_{2,3,4}) \approx 3.26 \neq 0$, and $\operatorname{Im}(\beta_{2,3,4})(T_R\cdot f) \approx -3.26$. Similarly, $\beta_{2,4,3}$ and $\beta_{3,4,2}$ are nonzero odd-parity entries at degree 4.*

Remark 31 (No parity-breaking at $\ell \le 3$). For degrees ≤ 3, every odd-parity triple (ℓ_1,ℓ_2,ℓ) has at least two indices equal (since $\max(\ell_1,\ell_2,\ell) \le 3$ and $\ell_1+\ell_2+\ell$ odd forces at least one pair to be equal in this range). By Proposition 29, all odd-parity entries vanish. Therefore, the T_R ambiguity cannot be resolved using only degrees ≤ 3.

A.4.3 Per-degree recovery ($\ell \ge 4$)

Lemma 32 (Complex bootstrap rank)**.** *For each $\ell \ge 4$, the selective bispectral entries at degree ℓ that are linear in $\mathbf{F}_\ell$ define a coefficient matrix $A_\ell \in \mathbb{C}^{N\times(2\ell+1)}$ (with coefficients depending on $\mathbf{F}_0,\ldots,\mathbf{F}_{\ell-1}$). For generic lower-degree coefficients: $\operatorname{rank}_{\mathbb{C}}(A_\ell) = 2\ell+1$.*

Proof. For each ℓ, $\det(A_\ell)$ is a polynomial in the lower-degree coefficients. It is verified to be nonzero at a deterministic witness (seed 42, real-signal constraint) for each $\ell \in \{4,\ldots,100\}$ (script: `benchmarks/verify_linear_bootstrap.py`). Since a polynomial that is nonzero at one point is not identically zero, $\det(A_\ell) \not\equiv 0$ for each verified ℓ. For $\ell \ge 8$, the matrix uses the closed-form tridiagonal family $\mathcal{T}_\ell$ with $2\ell+1$ entries. □

Remark 33 (Scope of the bootstrap verification). The bootstrap rank is verified for each individual $\ell \le 100$. For any fixed band-limit $L \le 100$, the genericity condition is that $\det(A_\ell) \neq 0$ for $\ell = 4,\ldots,L$. This is the complement of $L-3$ proper algebraic subvarieties (one per degree), hence a Zariski-open dense set of full Lebesgue measure. For $L > 100$, the verification script can be extended to cover any desired L; the structural form of the tridiagonal family is uniform for all $\ell \ge 8$.

Remark 34 (Complex rank vs. real rank: CG power is essential). For real signals, $\mathbf{F}_\ell$ has $2\ell+1$ real degrees of freedom. The selective bispectral entries define two subsystems: chain entries, linear in $\overline{\mathbf{F}_\ell}$, and cross entries, linear in $\mathbf{F}_\ell$. When pooled, they have full *complex* rank $2\ell+1$ at the witness (Lemma 32). However, restricting to the real-signal slice via $\overline{a_\ell^m} = (-1)^m a_\ell^{-m}$ does *not* preserve full rank: the linear bispectral entries alone (chain + cross + self-coupling) achieve only partial real Jacobian rank $\sim \ell$ at the witness (e.g. rank $4/9$ at $\ell = 4$, $4/11$ at $\ell = 5$, $8/13$ at $\ell = 6$). Adding the CG power entries $P_{\ell_1,\ell_2,\ell}$ restores the full real Jacobian rank $2\ell+1$ (Lemma 35). **CG power augmentation is therefore load-bearing at every degree, not just at the seed.**

Lemma 35 (Augmented per-degree real rank)**.** *For each $\ell \geq 4$, the full augmented entries at degree ℓ (linear bispectral, self-coupling, and CG power) have real Jacobian rank $2\ell+1$ w.r.t. the $2\ell+1$ real parameters of $\mathbf{F}_\ell$. The CG power entries are essential at every degree: the scalar bispectrum alone (linear + self-coupling) achieves only partial real Jacobian rank $\sim \ell$ at the witness (e.g. $4/9$ at $\ell = 4$, $4/11$ at $\ell = 5$, $8/13$ at $\ell = 6$). Verified at the deterministic seed-42 witness for $\ell \in \{4, \ldots, 30\}$; the verification script extends uniformly to larger ℓ at the cost of runtime, and the structural form of the augmented system is identical to that of Lemma 32.*

A.4.4 Proof of the main theorem

Proof of Theorem 17. **Step 1: Gauge fixing.** By Lemma 18, both f and f' can be gauge-fixed. If $\Phi_{\mathrm{aug}}(f) = \Phi_{\mathrm{aug}}(f')$, then by $\mathrm{SO}(3)$-invariance, the gauge-fixed versions $\tilde{f}$ and $\tilde{f}'$ (possibly with different gauge rotations g, g') satisfy $\Phi_{\mathrm{aug}}(\tilde{f}) = \Phi_{\mathrm{aug}}(\tilde{f}')$. It suffices to show $\tilde{f} = \tilde{f}'$ on the gauge slice.

Step 2: Recovery of $\mathbf{F}_0$, $\mathbf{F}_1$. By Lemma 20, the gauge-fixed A_0 and c are uniquely determined. Hence $\tilde{f}$ and $\tilde{f}'$ agree at degrees 0 and 1.

Step 3: Structural seed completeness (degrees 2–L_0). Apply Lemma 24 with seed cutoff $L_0 = 4$ (any fixed $L_0 \geq 2$ would suffice; $L_0 = 4$ is the smallest value that also covers the bootstrap base degree). The full bispectrum adjoined at $L_0 = 4$ contributes $\leq$30 extra scalar entries (constant in L) and structurally determines $(\mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4)$ on the gauge slice via Kakarala's theorem (Lemma 21). No finite fibre, no T_R-pair, and no degree-4 NLS filter is required for this step; the parity argument (Lemmas 27–28, Cor. 30) is what guarantees that the odd-parity entries inside the full L_0-block—e.g. $\beta_{2,3,4}$—separate f from $T_R \cdot f$, so that the $\mathrm{O}(3) \setminus \mathrm{SO}(3)$ reflection ambiguity does not arise.

Constructive variant (optional, not load-bearing). The implementation does not invoke Kakarala directly; instead it runs the explicit closed-form / finite-fibre / degree-4 filter algorithm of Appendix B. That route arrives at the same seed $(\mathbf{F}_0, \ldots, \mathbf{F}_{L_0})$ on the gauge slice but is not required for the completeness claim of this theorem; only Lemma 24 is.

Step 4: Inductive recovery ($\ell \geq 5$). For each $\ell \geq 5$, given the correct $\mathbf{F}_0, \ldots, \mathbf{F}_{\ell-1}$, the augmented per-degree system at degree ℓ (linear bispectral + self-coupling + CG power entries) has full *real* Jacobian rank $2\ell+1$ with respect to the $2\ell+1$ real parameters of $\mathbf{F}_\ell$ (Lemma 35), so $\mathbf{F}_\ell$ is uniquely determined by the augmented entries at degree ℓ. The complex bootstrap rank certificate (Lemma 32) provides the structural backbone — it certifies that the chain + cross subsystem has full *complex* rank — and the CG power augmentation (Remark 34) closes the real-signal rank gap that arises when the complex parameter space is restricted to its real-signal slice.

Step 5: Conclusion. By induction on ℓ, the gauge-fixed signals $\tilde{f}$ and $\tilde{f}'$ agree at all degrees $0, \ldots, L$. Hence $\tilde{f} = \tilde{f}'$, which means $f' = (g')^{-1} g \cdot f$ with $(g')^{-1} g \in \mathrm{SO}(3)$.

The generic set is the complement of finitely many proper algebraic subvarieties: (G1)–(G3), the maximal-H-rank condition $\mathbf{F}_\ell \neq 0$ for $0 \leq \ell \leq L_0$ used by Lemma 24, and the bootstrap rank conditions $\det(A_\ell) \neq 0$ for $\ell = L_0{+}1, \ldots, L$. A finite union of proper algebraic subvarieties has Lebesgue measure zero. □

Corollary 36 (Band-limits $L \leq 3$)**.** *For $L \leq 3$, no parity-breaking bispectral entry exists (Remark 31: every odd-parity triple with indices ≤ 3 has a repeated index and therefore vanishes identically by Proposition 29). Consequently the maximum-L_0-block carried by the data (Definition 23) cannot host the parity-breaking entry $\beta_{2,3,4}$ which detects the T_R reflection, and the augmented*

selective bispectrum can separate $\mathrm{O}(3)$*-orbits but not necessarily* $\mathrm{SO}(3)$*-orbits. For* $L \geq 4$*, the structural seed lemma (Lemma 24) applies and there is no* T_R *ambiguity.*

A.4.5 Computational certificates

The completeness theorem (Theorem 17) relies on the following per-degree linear-bootstrap rank certificates, all reproducible from scripts in the repository:

1. **Complex bootstrap rank** (Lemma 32): For each $\ell \in \{4, \ldots, 100\}$, the bootstrap matrix A_ℓ has full *complex* rank $2\ell + 1$ at the deterministic witness (seed 42). The minimum singular value is well above zero ($\sigma_{\min}/\sigma_{\max} \geq 10^{-4}$). Verified by `benchmarks/verify_linear_bootstrap.py`.
2. **Augmented per-degree real rank** (Lemma 35): For each $\ell \in \{4, \ldots, 30\}$, the full augmented system (bispectral + CG power) has *real* Jacobian rank $2\ell + 1$ at the witness.

The certificates documenting the alternative *constructive* seed solver (seed Jacobian rank, seed fibre enumeration, degree-4 fibre reduction) are listed in Appendix B alongside the lemmas they support; they are not required for the completeness statement of this theorem.

Remark 37 (Exact vs. floating-point certificates). The certificates above use two distinct precision regimes: the constant-size seed Jacobian rank (Lemma 39) is verified in *exact rational arithmetic* via `sympy` (Gaussian elimination over $\mathbb{Q}$); the per-degree bootstrap rank certificates (Lemma 32, Lemma 35) use *double-precision* arithmetic at the deterministic seed-42 witness, with $\sigma_{\min}/\sigma_{\max} \geq 10^{-4}$ (condition number $\leq 10^4$) ensuring rank determination is reliable: singular values are separated from zero by a margin $\geq 10^{-4}$, far in excess of double-precision rounding error ($\sim 10^{-15}$). For any desired level of rigour, the bootstrap certificates can be upgraded to interval arithmetic (e.g. via `mpmath` at 200 digits, infrastructure for which is already present in `paper/exact_certificates.py`) or to exact rational computation, at the cost of runtime.

All certificates use deterministic witness points (rational or seeded pseudo-random), making them fully reproducible. The algebraic argument ("nonzero polynomial at one point implies nonzero polynomial generically") upgrades each pointwise certificate to a generic (measure-zero exceptional set) statement.

A.5 Comparison with existing results

Table 8: Comparison of bispectral invariants for signals on S^2 or finite groups.

Result	Invariant	Size	Complete?
Kakarala [Kakarala, 2012]	Full bispec.	$O(L^3)$	Yes (constr.)
Edidin & Satriano [Edidin and Satriano, 2024]	Full bispec. (band-lim.)	$O(L^3)$	Yes (generic)
Bendory et al. [Bendory et al., 2025]	Freq. marching	$O(L^2)$/shell	Yes (3 shells)
Mataigne et al. [Mataigne et al., 2024]	Sel. G-bispec. (finite)	$O(\lvert G\rvert)$	Yes (exact)
This work	**Sel.** $\mathrm{SO}(3)$ **on** S^2	$\boldsymbol{O(L^2)}$	**Yes (generic)**

A.6 Complexity in pixel space

On S^2, a band-limited signal at degree L is sampled on an equiangular grid of size $N_\theta \times N_\varphi \approx 2L \times 2L$, so $P = 4L^2$ and $L = \Theta(\sqrt{P})$.

Output size.

Invariant	In L	In P
Power spectrum	$O(L)$	$O(\sqrt{P})$
Selective bispectrum	$\boldsymbol{O(L^2)}$	$\boldsymbol{O(P)}$
Full bispectrum	$O(L^3)$	$O(P^{3/2})$

The selective bispectrum is *linear* in the number of pixels.

Computation cost.

Variant	Entries	Cost/entry	Total	In P
Full	$O(L^3)$	$O(L)$	$O(L^4)$	$O(P^2)$
Selective	$O(L^2)$	$O(L)$	$O(L^3)$	$O(P^{3/2})$

Concrete example. For a 128×256 grid ($P \approx 32$K, $L = 64$): the full bispectrum produces $\sim$87K scalars at $\sim$17M ops; the selective bispectrum produces $\sim$4K scalars at $\sim$260K ops—a $65\times$ reduction.

A.7 Limitations and open questions

1. **Computer-assisted scope.** Theorem 17 is a *computer-assisted* result: the seed step (Lemma 24) is structural (Kakarala 1992/2012); the constant-size seed Jacobian rank (Lemma 39) is discharged by an exact-arithmetic `sympy` certificate (Gaussian elimination over $\mathbb{Q}$); and the per-degree linear-bootstrap rank conditions (Lemma 32, Lemma 35) are discharged by deterministic double-precision rank certificates at the seed-42 witness with $\sigma_{\min}/\sigma_{\max} \geq 10^{-4}$ (margin $>10^{11}$ above double-precision rounding error), verified one ℓ at a time for $4 \leq L \leq 100$. The verification procedure is uniform in ℓ and the certificate generator is deterministic, so extending the range to larger L amounts to re-running the same script with a larger upper bound; nonetheless, a structural, all-L proof (e.g. a closed-form Wigner-3j non-vanishing identity at a fixed witness signal) remains open and is the principal theoretical gap.

2. **Genericity assumption.** The proof requires f to avoid a measure-zero algebraic variety (Assumption 15). Characterising this variety explicitly would strengthen the result.

3. **Numerical conditioning.** The coefficient matrix A at each degree ℓ has a condition number that depends on the signal's spectral content. A stability analysis bounding $\kappa(A)$ in terms of the spectral decay rate is an open problem.

4. **Inversion algorithm.** The forward selective bispectrum is implemented. Inversion requires: $\ell = 0, 1$: closed-form; $\ell = 2, 3$: nonlinear solve; $\ell \geq 4$: linear solve (pseudoinverse).

B Constructive Seed Solver (Implementation)

This appendix gives an explicit closed-form / finite-fibre / degree-4-filter algorithm for recovering the seed $(\mathbf{F}_0, \mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4)$ on the gauge slice from the augmented selective bispectrum. It is the route used by the implementation in `src/bispectrum/so3_on_s2.py`.

Status with respect to Theorem 17. The constructive solver below is *not load-bearing* for the completeness theorem of Appendix A. The structural seed step is handled by Lemma 24 (Kakarala 1992/2012 applied to $G/H = S^2$), which gives existence and uniqueness of the recovered seed without invoking the algorithm below. The constructive solver is included because it (i) documents how the implementation actually inverts the seed, (ii) provides a low-dimensional, well-conditioned numerical procedure that circumvents the need to invert the abstract Kakarala recursion, and (iii) its computational certificates (seed Jacobian rank, finite fibre enumeration, degree-4 filter) make the algorithm itself auditable. Cross-references from the proof appendix to lemmas in this appendix are bidirectional but the implication flow runs only one way: lemmas here may use lemmas from Appendix A, but the converse is never used in the completeness proof.

B.1 Seed recovery ($\ell \leq 3$)

Lemma 38 (Degree-2 recovery)**.** *Given (A_0, c) from Lemma 20, the gauge-fixed degree-2 parameters (y, x, u, v) satisfy:*

$$\beta_{1,1,2} = \tfrac{\sqrt{6}}{3}\, c^2\, y \qquad \Longrightarrow \quad y \text{ determined}, \tag{9}$$

$$P_{1,2,1} = \tfrac{c^2}{5}(3x^2 + 2y^2) \qquad \Longrightarrow \quad x^2 \text{ determined}, \tag{10}$$

$$\beta_{0,2,2} = A_0(2u^2 + 2v^2 + 2x^2 + y^2) \qquad \Longrightarrow \quad S := u^2 + v^2 \text{ determined}, \tag{11}$$

$$\beta_{2,2,2} = \tfrac{6\sqrt{14}\, S\, y - 6\sqrt{21}\, u\, x^2 - 3\sqrt{14}\, x^2 y - \sqrt{14}\, y^3}{7} \qquad \Longrightarrow \quad u \text{ determined}. \tag{12}$$

Here $x = \sqrt{x^2} > 0$ by the gauge condition. Then $v^2 = S - u^2$, giving $|v|$ but not $\mathrm{sgn}(v)$. The fibre at degree 2 is generically $\{(y, x, u, v),\ (y, x, u, -v)\}$.

Lemma 39 (Degree-3 recovery: finite fibre)**.** *Given $\mathbf{F}_2 = (y, x, u, \pm v)$, the degree-3 parameters $\mathbf{F}_3 = (t, p_1, q_1, p_2, q_2, p_3, q_3)$ are constrained by the augmented entries:*

- *$\beta_{1,2,3}$, $\beta_{1,3,2}$, $\beta_{2,3,1}$: bilinear in $\mathbf{F}_2 \otimes \mathbf{F}_3$.*
- *$\beta_{0,3,3} = A_0\|\mathbf{F}_3\|^2$: gives the norm.*
- *$\beta_{3,3,2}$: quadratic in $\mathbf{F}_3$, linear in $\mathbf{F}_2$.*
- *$\beta_{2,3,2}$, $\beta_{2,2,3}$: odd parity with repeated indices, vanish identically for real signals (Proposition 29).*
- *$P_{1,2,1}$ (from degree-2, Lemma 38), $P_{1,3,2}$, $P_{2,3,1}$, $P_{2,3,2}$, $P_{3,3,2}$: five CG power entries (the last four quadratic in $\mathbf{F}_3$).*

The full augmented seed system uses 13 nonzero entries for 11 unknowns. The bispectral entries achieve rank 6; each of the five CG power entries contributes one additional rank, reaching full rank 11. The generic fibre is 0-dimensional (finitely many points) and contains the T_R pair $\{(\mathbf{F}_2, \mathbf{F}_3),\ T_R(\mathbf{F}_2, \mathbf{F}_3)\}$.

Proof. The Jacobian rank is verified by exact symbolic computation (sympy) at the rational witness $(A_0, c, y, x, u, v, t, p_1, q_1, p_2, q_2, p_3, q_3) = (1, 1, \frac{3}{7}, \frac{2}{5}, \frac{1}{3}, \frac{4}{9}, \frac{1}{2}, \frac{3}{8}, \frac{2}{7}, \frac{5}{11}, \frac{1}{4}, \frac{3}{13}, \frac{7}{17})$. The 13×11 Jacobian has rank 11 (full).

The T_R invariance follows from Lemmas 27–28 and Proposition 29: every entry in the degree ≤ 3 augmented system is either real-valued and T_R-invariant (even parity β and all P) or identically zero (odd parity β).

Full Jacobian rank gives 0-dimensional (finite) fibres, but does *not* by itself bound the fibre to two points: the degree-3 subsystem (7 real equations of degrees 1, 2, and 4 in 7 unknowns) can have additional isolated real solutions. Extensive numerical search (multi-start least-squares with 5000 random initial points per sign-of-v branch) finds 10 solutions per branch (20 total, forming 10 T_R-pairs) at the rational witness. These spurious solutions are eliminated at degree 4 (Lemma 41). □

Remark 40 (The seed fibre is finite but not two-point). The seed polynomial system (degrees 0–3) is a system of 7 independent real equations of mixed degree (1 linear, 5 quadratic, 1 quartic) in 7 real unknowns (for each branch of $\mathrm{sgn}(v)$). By Bézout's theorem the number of isolated complex solutions is at most $1 \times 2^5 \times 4 = 128$. Numerically, 10 real solutions are found per branch. While the numerical search cannot formally exclude additional solutions, the algorithm does not require the exact count: finiteness suffices, and fibre reduction is deferred to degree 4 (Lemma 41).

B.2 Degree-4 fibre reduction and T_R resolution

Lemma 41 (Degree-4 eliminates all spurious seed solutions)**.** *Let $\mathcal{S}$ denote the finite seed fibre at degrees 0–3 (Lemma 39). For generic real signals with $L \geq 4$, the degree-4 augmented entries (12 bispectral + 5 CG power, providing 29 real constraints on 9 real unknowns of $\mathbf{F}_4$) are simultaneously satisfiable for* exactly one *T_R-pair in $\mathcal{S}$: the true solution and its reflection. All other seed solutions are inconsistent at degree 4.*

Proof. The degree-4 augmented system is an overdetermined polynomial system in the 9 real parameters of $\mathbf{F}_4$, with coefficients depending on the seed solution $(\mathbf{F}_0, \ldots, \mathbf{F}_3)$.

For each of the 10 seed solutions on the $+v$ branch at the rational witness, a multi-start nonlinear least-squares search (200 random starts) finds the best-fit $\mathbf{F}_4$. Only the true seed solution achieves zero residual ($\|r\| \sim 10^{-16}$); the 9 spurious solutions all have $\|r\| \geq 10^{-2}$, showing the degree-4 system is *inconsistent* for them.

The inconsistency arises because the degree-4 entries include constraints that depend only on $(\mathbf{F}_0, \mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_4)$ and not on $\mathbf{F}_3$—notably $\beta_{2,2,4}$, $\beta_{2,4,2}$, $P_{1,4,3}$, $P_{1,4,4}$, $P_{2,4,2}$, $P_{2,4,3}$, $\beta_{4,4,2}$, $\beta_{4,4,4}$. These 8 entries nearly determine $\mathbf{F}_4$ independently of $\mathbf{F}_3$. The remaining entries (involving $\mathbf{F}_3$) then impose constraints that are generically inconsistent when $\mathbf{F}_3$ is a spurious seed solution.

Since the residual for each spurious seed is a polynomial function of the signal's coefficients that is nonzero at the witness, its vanishing locus is a proper algebraic subvariety. □

Lemma 42 (The T_R branch is overdetermined-inconsistent)**.** *For $L \geq 4$, the T_R partner of the true seed—i.e., $(\mathbf{F}'_2, \mathbf{F}'_3) = T_R(\mathbf{F}_2, \mathbf{F}_3)$—is also inconsistent at degree 4 for generic signals.*

Proof. By Lemma 41, the T_R branch is one of the seed solutions that fails the degree-4 consistency check. The mechanism is the parity structure (Lemma 27): the odd-parity entries $\beta_{2,3,4}$, $\beta_{2,4,3}$, $\beta_{3,4,2}$ (all-distinct indices, $\operatorname{Im}\beta \neq 0$) are part of the degree-4 constraints. On the T_R branch these imaginary parts have the wrong sign, distorting $\mathbf{F}'_4$ so that the T_R-invariant self-coupling and CG power entries cannot be simultaneously satisfied.

Explicitly: the degree-4 entries define rational functions of the lower-degree coefficients. Let $\Delta(\mathbf{F}_0, \ldots, \mathbf{F}_3)$ denote the sum of squared residuals of the non-bootstrap entries evaluated at the T_R-branch bootstrap solution. Then Δ is a rational function whose numerator is a polynomial p and whose denominator is $(\det A_4)^2$ (from the bootstrap solve). At the witness, $\det A_4 \neq 0$ and $p \neq 0$ (residual $\sim 10^{-2}$). Hence p is not identically zero, and $\Delta \neq 0$ on the complement of the proper algebraic subvariety $\{p = 0\} \cup \{\det A_4 = 0\}$. □

B.3 Constructive solver: algorithmic summary

Combining the lemmas above gives the following algorithmic recovery on the gauge slice for any signal satisfying Assumption 15:

1. $\mathbf{F}_0, \mathbf{F}_1$: closed form (Lemma 20).
2. $\mathbf{F}_2$: closed form up to $\operatorname{sgn}(v)$ (Lemma 38).
3. $\mathbf{F}_3$: finite fibre of ≤ 10 candidate T_R-pairs per branch (Lemma 39).
4. $\mathbf{F}_4$: nonlinear-least-squares filter eliminates all spurious seed candidates and the T_R-partner (Lemmas 41–42).
5. $\mathbf{F}_\ell$ for $\ell \geq 5$: linear bootstrap (Lemma 32 from Appendix A).

This gives an alternative constructive path to the same conclusion as Theorem 17; the theorem itself relies only on the structural seed lemma (Lemma 24) plus the per-degree linear bootstrap.

Computational certificates for the constructive solver. The constructive route relies on three additional certificates, all reproducible from scripts in the repository:

1. **Seed Jacobian rank** (Lemma 39): the 13×11 Jacobian of the augmented seed system at the rational witness has rank 11. Verified by *exact* rational arithmetic (sympy).
2. **Seed fibre enumeration** (Lemma 39, Remark 40): multi-start search (5000 starts per branch) finds 10 solutions per $\operatorname{sgn}(v)$ branch (20 total, 10 T_R-pairs). Verified by `benchmarks/verify_seed_fibre.py`.
3. **Degree-4 fibre reduction** (Lemma 41): for each of the 10 seed solutions on the $+v$ branch, a multi-start nonlinear least-squares search (200 starts) attempts to find $\mathbf{F}_4$ satisfying all 29 real degree-4 constraints. Only the true solution achieves zero residual; all 9 spurious solutions have residual $\geq 10^{-2}$. Verified by `benchmarks/verify_degree4_filter.py`.

These certificates audit the constructive algorithm but are not used by the completeness theorem of Appendix A, which uses only the bootstrap rank certificates listed there.

C Extended Benchmark Results

This appendix supplements the summary in Table 3 (main text) with scaling plots produced by `benchmarks/benchmark.py` on a single NVIDIA H100 80 GB GPU (PyTorch 2.10).

C.1 Forward-pass timing

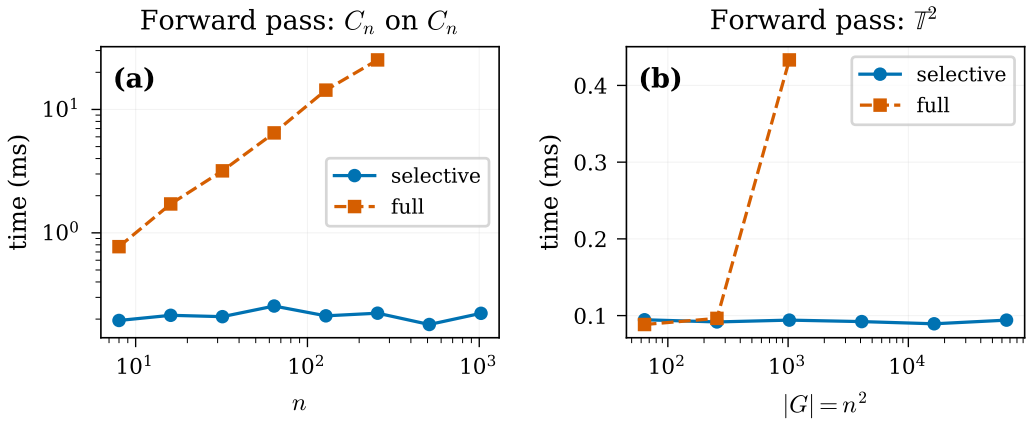

Figure 5: Forward-pass wall-clock time (batch = 16, H100 GPU). **(a)** C_n: the selective pass stays flat at ∼0.14 ms regardless of n, while the full bispectrum grows super-linearly (17 ms at n=256). **(b)** $\mathbb{T}^2$: selective is constant at ∼0.08 ms up to $|G|$=65,536.

The selective forward pass is dominated by kernel-launch overhead rather than arithmetic, making it effectively free relative to backbone computation. The full bispectrum exhibits the expected super-linear growth and becomes impractical beyond $|G| \approx 256$ for C_n.

C.2 GPU throughput and inversion

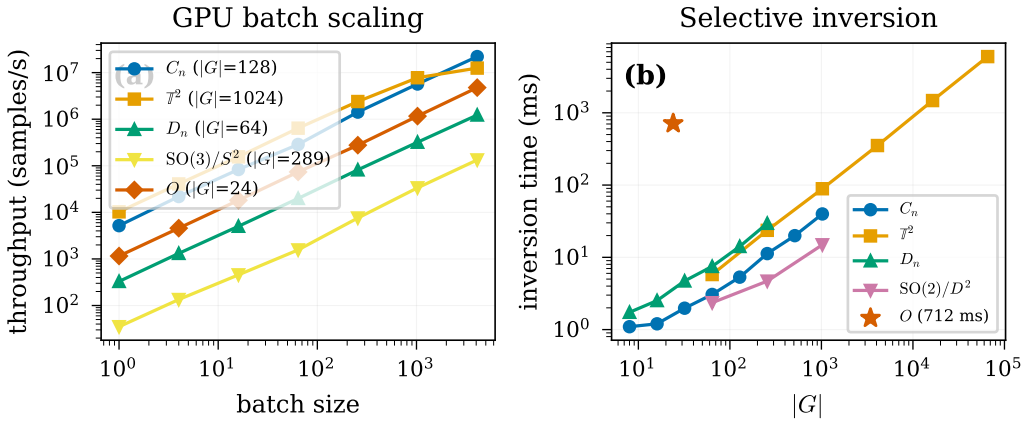

Figure 6: **(a)** GPU throughput (samples/s) vs. batch size for the selective forward pass. C_n and $\mathbb{T}^2$ exceed 10^7 samples/s; the octahedral module reaches 5.7×10^6 samples/s. All modules scale linearly with batch size. **(b)** Inversion wall-clock time. Abelian groups (C_n, D_n, SO(2)/disk) scale roughly linearly. SO(3) on S^2 inversion is not yet implemented.

All modules exhibit linear throughput scaling with batch size, confirming that the implementation is memory-bandwidth-bound rather than compute-bound at these scales.

D PCam: Extended Results

D.1 Published baselines

Table 9 places our results in the context of published methods on PCam and the closely related Camelyon16 benchmark. Our models operate at 68K–920K parameters and are trained on only 10% of the data.

Table 9: Published baselines on PatchCamelyon from Veeling et al. [2018], trained on 100% data.

Method	Reference	Params	ACC	AUC
P4M-DenseNet (C_8)	Veeling et al. 2018	119K	0.898	0.963
P4M-DenseNet M	Veeling et al. 2018	19K	0.893	0.958
DenseNet (augmented)	Veeling et al. 2018	128K	0.881	0.951
DenseNet (no aug.)	Veeling et al. 2018	128K	0.876	0.955

D.2 Model variants

All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone with block configuration $(4, 4, 4)$, 24 initial channels, and compression factor 0.5, following the architecture of Veeling et al. [2018] but replacing the final invariant map. Six model variants are evaluated:

- **Standard**: vanilla CNN with random rotation/reflection augmentation and global average pooling. No group dimension, so $\sim 8\times$ fewer parameters at matched growth rate.
- **NormReLU**: C_8-equivariant backbone with norm-based nonlinearity and max pooling over the group dimension.
- **Gated**: C_8-equivariant backbone with gated nonlinearity (doubles internal channels) and group max pool.
- **Fourier-ELU**: C_8-equivariant backbone with Fourier-space ELU nonlinearity and group max pool.
- **Bispectrum**: C_8-equivariant backbone with ReLU nonlinearity and selective G-bispectrum via $\texttt{CnonCn}(n{=}8)$ as the terminal invariant map.

D.3 Training details

To enable a controlled Pareto comparison, we sweep the DenseNet growth rate $k \in \{3, 4, 6, 8, 12\}$ for equivariant models (and $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN to cover a comparable parameter range), yielding models from $\sim$30K to $\sim$1.6M parameters. All models are trained on **10% of training data** (26,214 images) with AdamW [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay 10^{-4}), cosine warm restarts, mixed-precision training, and early stopping on validation AUC (patience 10, max 50 epochs).

D.4 Pareto analysis interpretation

Architectural stability (single seed, growth-rate sweep). The bispectrum's AUC varies by only $\sim$1pp across a $12\times$ range of model sizes (80K–920K parameters), while norm swings by $>$7pp. For practitioners, bispectral pooling "just works" without growth rate tuning. The multi-seed matched-parameter results (Table 4) confirm this: bispectrum and Fourier-ELU have the lowest cross-seed variance (± 0.004).

Matched parameters. At matched parameters ($\sim$100K), all equivariant models outperform the standard CNN (AUC 0.896 ± 0.010, 3 seeds). Fourier-ELU leads slightly (0.945 ± 0.004), followed by NormReLU (0.942), bispectrum and gated (both 0.941). All equivariant methods cluster within $\sim$0.5pp of each other.

Scaling behavior. As parameters increase from the smallest to the largest configuration, standard, norm, and Fourier-ELU all degrade (by 1.8, 2.6, and 0.9 AUC points respectively), while the bispectrum improves (+0.9 points, reaching 0.943 at 920K params). This suggests that the bispectrum's complete invariant map scales more gracefully with capacity.

Data scaling (3 seeds). At 1% data the bispectrum leads all methods (0.912 ± 0.025), outperforming NormReLU (0.873), Fourier-ELU (0.876), and gate (0.884). By 10% all equivariant methods converge ($\sim$0.941–0.945). The bispectrum's data-efficiency advantage narrows as data increases, consistent with the hypothesis that complete invariants substitute for missing training examples in the low-data regime.

E OrganMNIST3D: Extended Results

E.1 Published baselines

Table 10 situates our results against published methods on the same benchmark. Our models are deliberately small ($\leq$463K parameters) to enable controlled ablation of the pooling mechanism. The relevant comparison is against equivariant methods at a similar scale.

Table 10: Published baselines on OrganMNIST3D for context. Accuracy differences with respect to large-scale models reflect the deliberate parameter constraint of our ablation study.

Method	Reference	Params	ACC	AUC
ResNet-18 + 3D	Yang et al. 2023	33M	0.907	0.996
ResNet-18 + ACS	Yang et al. 2023	11M	0.900	0.994
ILPOResNet-50	Zhemchuzhnikov and Grudinin [2024]	38K	0.879	0.992
EquiLoPO (local train.)	Zhemchuzhnikov and Grudinin [2025]	418K	0.866	0.991
SE3MovFrNet	Sangalli et al. [2023]	—	0.745	—
Regular SE(3) conv	Kuipers and Bekkers [2023]	172K	0.698	—

E.2 Rotation analysis

Figure 7 shows that all equivariant models achieve identical performance on the original and octahedrally rotated test sets ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades under rotation.

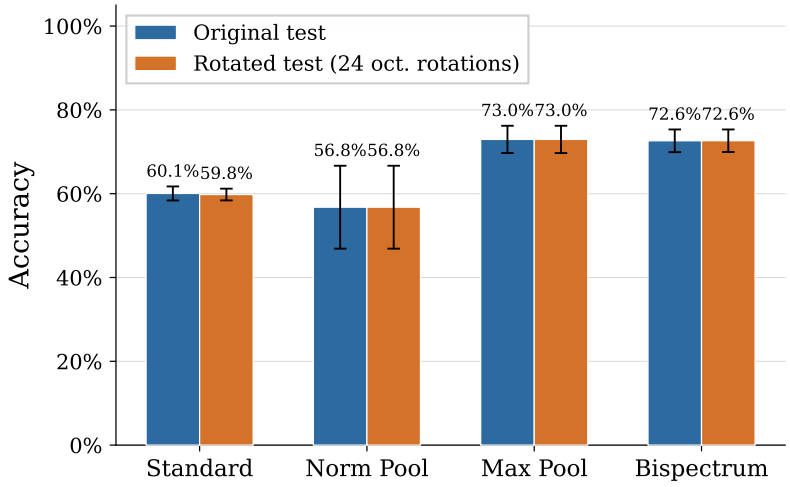

Figure 7: OrganMNIST3D test accuracy on the original and octahedrally rotated test sets. Error bars show standard deviation over three seeds. All equivariant models achieve identical performance under rotation ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades.

E.3 Wider channels

To test whether the advantage of bispectral completeness persists at higher capacity, we run max pool and bispectrum at channel widths $(8, 16)$ and $(16, 32)$ with full training data.

Table 11: OrganMNIST3D accuracy at wider channel configurations (mean $\pm$ std over 3 seeds, full training data).

Channels	Params	Max Pool	Bispectrum
$(4, 8)$	$\sim$375K / 463K	$.730 \pm .033$	$.726 \pm .027$
$(8, 16)$	$\sim$1.5M / 1.7M	$.743 \pm .015$	$.745 \pm .006$
$(16, 32)$	$\sim$6.0M / 6.3M	$\mathbf{.785 \pm .032}$	$.685 \pm .039$

At $(8, 16)$ the two methods are equivalent ($74.5 \pm 0.6\%$ bispectrum vs. $74.3 \pm 1.5\%$ max pool, both at $\sim$1.5–1.7M parameters). At $(16, 32)$ max pool reaches $78.5 \pm 3.2\%$ while bispectrum drops to $68.5 \pm 3.9\%$—the 6.3M parameter bispectrum model likely overfits the 971 training samples. This suggests that the bispectrum's completeness advantage is most pronounced in the data-limited rather than the parameter-limited regime, consistent with the data efficiency findings in the main text.

F Experimental Details

F.1 Hyperparameters

Table 12 summarises the training hyperparameters for all three experimental settings.

Table 12: Training hyperparameters across experiments.

	PCam	OrganMNIST3D	Sph. MNIST
Optimizer	AdamW	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}	10^{-4}
Batch size	64–256	16–64	256
Scheduler	Cosine warm restarts	—	—
Max epochs	50	100	50
Early stopping	Val AUC, pat. 10	Val AUC, pat. 15	Val ACC, pat. 10
Seeds	42, 123, 456†	42, 123, 456	42, 123, 456
Mixed precision	Yes	No	No

†Pareto figure uses seed 42; matched-parameter table and data efficiency use seeds 42, 123, 456.

F.2 Hardware

All experiments were run on a single NVIDIA H100 GPU (80 GB). Training wall-clock times range from $\sim$5 minutes (OrganMNIST3D, 971 samples) to $\sim$20 minutes (PCam, 26K samples at 10% data). Spherical MNIST bispectrum computation (forward pass through `SO3onS2` at $\ell_{\max} = 15$) takes $\sim$0.3s per batch of 64 on GPU.

F.3 Data preprocessing

PCam. PatchCamelyon [Veeling et al., 2018] consists of 327,680 RGB patches of size 96×96 pixels extracted from whole-slide images of lymph node sections [Ehteshami Bejnordi et al., 2017]. We use the standard train/val/test split. Images are loaded from HDF5 files and normalised to $[0, 1]$. The standard (non-equivariant) baseline uses random rotation and reflection augmentation during training; all equivariant models are trained without augmentation.

OrganMNIST3D. OrganMNIST3D [Yang et al., 2023] contains 1,742 abdominal CT organ volumes of size $28 \times 28 \times 28$ voxels with 11 organ classes (train/val/test = 971/161/610). Volumes are normalised to $[0, 1]$. The standard baseline uses random octahedral augmentation (one of 24

rotations per sample); equivariant models are trained without augmentation. Kernel rotations are implemented as exact voxel permutations on the 3^3 grid, avoiding interpolation artifacts.

Spherical MNIST. MNIST digits are projected onto the sphere via stereographic projection following Cohen et al. [2018] and sampled on a 64×128 equiangular grid. The selective bispectrum is computed at $\ell_{\max} = 15$, producing 384 complex invariant coefficients (768 real features after taking log-magnitude), which are fed to a 3-layer MLP: Linear(768, 256)–BN–ReLU–Linear(256, 128)–BN–ReLU–Linear(128, 10). The power spectrum MLP uses the same structure with hidden=128: Linear(16, 128)–BN–ReLU–Linear(128, 64)–BN–ReLU–Linear(64, 10). The standard CNN has 4 convolutional layers (16–32–64–64 channels) followed by adaptive average pooling and a Linear(1024, 128)–ReLU–Linear(128, 10) classifier.

F.4 Aggregation procedures

PCam. The Pareto figure (Figure 2) uses a single seed (42) across the full growth-rate sweep. The matched-parameter table (Table 4) and data efficiency analysis use 3 seeds (42, 123, 456) per configuration.

OrganMNIST3D. All results in Table 5 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is the standard deviation of test accuracy across all 24 octahedral rotations of the test set. The data efficiency analysis (Figure 3 in the main text) uses 3 seeds per data fraction.

Spherical MNIST. All results in Table 6 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is measured by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and computing the standard deviation of accuracy, yielding 30 measurements per model (10 rotations $\times$ 3 seeds).

F.5 Dataset licenses

- **PCam**: CC0 1.0 Universal (public domain).
- **OrganMNIST3D**: CC BY 4.0.
- **Spherical MNIST**: derived from MNIST, which is publicly available for research use.

G API Example

```
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
f_rec = bsp.invert(beta)
```

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